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# Multiwavelength Analysis of Six Luminous, Fast Blue Optical Transients

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## ABSTRACT

We present multiwavelength observations and analysis of six luminous fast blue optical transients (LFBOTs) discovered in Zwicky Transient Facility (ZTF) survey data **taken** $\lesssim 150$–**400 days after the transient's discovery**. We identified these LFBOTs from their fast light-curve evolution ($t_{1/2} \leq 12\,\mathrm{d}$), blue colors at peak brightness ($g - r \leq -0.\mathbf{2}\,\mathrm{mag}$), a visible host galaxy, high optical

luminosity ($M_g < -20$), and an X-ray **and/or** radio detection. With the exception of AT2024aehp (ZTF24abygbss), these transients exhibit peaks in their 10 GHz radio light curves at $t_{\rm rest} \approx 50$–100 d, with peak radio luminosities ranging from $10^{38}$–$10^{40}$ erg s$^{-1}$. Modeling the radio emission as synchrotron radiation indicates a fast ($v = 0.1$–$0.3c$) shock in a dense ($n_e \approx 10^3$–$10^4$ cm$^{-3}$) medium. The X-ray emission varies by $\approx 2$ orders of magnitude in luminosity ($10^{42}$–$10^{44}$ erg s$^{-1}$) at $t_{\rm rest} \sim 20$ d. Analysis of the host-galaxy photometry and spectroscopy for each transient shows that they are predominantly nonnuclear (a few kpc offset) with star-forming host galaxies of stellar masses $10^9$–$10^{11}$ $M_\odot$. Unlike all other LFBOTs to date, AT2024aehp exhibited a luminous ($M < -19$ mag) plateau in the optical light curve **and its** 6–15 GHz radio emission brightened by over an order of magnitude from $t_{\rm rest} \approx 70$ d to $t_{\rm rest} \approx 130$ d. The mostly consistent radio behavior between **these** LFBOTs implies a similar circumburst medium, leading us to prefer a progenitor scenario in which mass is lost in a consistent way shortly prior to the terminal event.

## 1. INTRODUCTION

The advent of high-cadence optical surveys such as the Asteroid Terrestrial-impact Last Alert System (ATLAS; Tonry et al. 2018) and the Zwicky Transient Facility (ZTF; Bellm et al. 2019, Graham et al. 2019, Dekany et al. 2020) has led to the discovery of new classes of extragalactic transients. Some **of these** transients are referred to as fast blue optical transients (FBOTs), most famously AT2018cow (e.g., Prentice et al. 2018). FBOTs are typically defined as transients with blue colors ($g - r \leq -0.2$ mag), rapid time evolution (< 12 days above half-maximum brightness), and extragalactic nature (Drout et al. 2014; Pursiainen et al. 2018; Inserra 2019).

In this paper, we focus on a particular class of FBOTs with luminous X-ray and/or radio emission, dubbed "AT2018cow-like" transients (Ho et al. 2023b) or "luminous fast blue optical transients" (LFBOTs; Metzger 2022; Klencki & Metzger 2025). There are currently eight published transients that fall under this classification (Table 1). It is theorized that the peak of their optical light curves results from shock breakout from a dense medium (Ofek et al. 2010; Pursiainen et al. 2018) or from a jet cocoon (Gottlieb et al. 2022). The radio emission is found to be consistent with synchrotron radiation emanating from the mildly relativistic shock (Margutti et al. 2019; Coppejans et al. 2020; Ho et al. 2020a). The X-ray radiation is suggested to come from a central X-ray source, separate from the synchrotron radiation that powers the radio (Margutti et al. 2019; Ho et al. 2019). For objects with high-cadence X-ray observations (AT2018cow, AT2020mrf, AT2022tsd, and AT2024wpp) it was found that the X-rays had fast variability on a timescale of minutes to days (Rivera Sandoval et al. 2018; Margutti et al. 2019; Yao et al. 2022; Ho et al. 2023c; Nayana et al. 2025; Perley et al. 2026).

Based on the first few LFBOTs discovered, Metzger (2022) argued that progenitor models must reconcile two key characteristics: a dense surrounding medium inferred from the synchrotron radiation, and a long-lived central engine (accreting compact object) inferred from nonthermal variable X-ray emission as well as minutes-duration optical flares (Ho et al. 2023c). Some propositions include tidal disruption events (TDEs) involving an intermediate-mass black hole (Perley et al. 2019; Kuin et al. 2019; Gutiérrez et al. 2024), the direct collapse (failed explosion) of a supergiant star (Margutti et al. 2019; Perley et al. 2019; Chrimes et al. 2025), and the tidal disruption of a star by a stellar-mass black hole resulting from a merger (Metzger 2022; Klencki & Metzger 2025) or random interactions in a cluster (Kremer et al. 2021, 2023).

**While progenitor theories rely on the characteristics that are similar across LFBOTs, it is also important to take note of these transients' observational diversity. Quite a few of these LFBOTs exhibit unique behavior—AT2018cow remained UV-bright for several years (Inkenhaag et al. 2025), AT2020xnd was more X-ray luminous than other LFBOTs by orders of magnitude (Yao et al. 2022), AT2022tsd has minute-timescale optical flares (Ho et al. 2023c), etc. The cause of these differences is also uncertain, though some have proposed that viewing angle effects may explain the range in X-ray luminosity (Chrimes et al. 2024a; Nayana et al. 2025).**

**In all, our understanding of LFBOTs is incomplete, in large part due to our small LFBOT sample size. This makes it all the more crucial to increase the number of AT2018cow-like objects that are well-characterized by thorough multi-wavelength observations.** Unfortunately, the rarity of these transients and the recency of the development of optical surveys like ZTF has limited the discovery rate.

**In this paper, we present five more LFBOTs with multiwavelength follow-up observations, as well as further radio observations on a (AT2023fhn), increasing the sample size by over 50%. These LFBOTs were first identified as candidates in the ZTF alert stream on the basis of their color, optical light curve rise or decay rate, and the presence of a host galaxy within a few arcseconds of the transient. We then confirm them as LFBOTs through spectroscopy to measure their redshift and therefore luminosity, and by a detection in the X-rays or radio. The discovery and classification procedure of these six LFBOTs are detailed in Section A. After detection, we conduct broad multiwavelength follow-up campaigns that can extend up to years past their initial discovery. This includes radio and millimeter observations to constrain both sides of a broken power-law spectral energy distribution (SED) and the evolution of the SED's peak, X-ray evolutions to get X-ray detections and X-ray flux variability, and optical follow-up to observe the light curve past the ZTF sensitivity limit. In this paper, we focus on early-time observations $t_{\rm obs} \lesssim 400\,$d. As LFBOTs were only discovered within the past few years, the late-time picture is still developing, with ongoing efforts for long-term optical follow-up (Chen et al. 2023; Inkenhaag et al. 2025).**

This paper presents optical photometry, optical spectroscopy, and observations at X-ray, radio, and millimeter wavelengths. Section 2 describes the observations and data reduction. **In Section 3 we model the radio emission as self-absorbed synchrotron radiation. In Section 4 we analyze the offsets from each LFBOT to its host galaxy, and each host galaxy's properties. We use methods that are similar as possible to those done in previous LFBOT papers in order to make effective comparisons. We present our discussion and conclusions in Section 5, which also includes suggestions for future LFBOT studies.**

Throughout this paper, we assume a flat ΛCDM cosmology with $\mathrm{H}_0 = 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_m = 0.315$ (Planck Collaboration et al. 2020). All usages of $t_{\rm obs}$ refer to the time after the optical light curve peak in the observer frame, while $t_{\rm rest}$ refers to this time in the rest frame. All dates are in UTC and all magnitudes are in AB.

## 2. OBSERVATIONS

The basic characteristics for all LFBOTs are listed in Table 1. The discovery details for AT2022abfc, AT2023fhn, AT2023hkw, AT2023vth, AT2024qfm, and AT2024aehp are given in Appendix A. Here we provide details on our multiwavelength follow-up observations.

### 2.1. *Optical Photometry*

After each LFBOT was identified in ZTF photometry (outlined in Appendix A), we obtained follow-up optical photometry using the Liverpool Telescope (LT; Steele et al. 2004) and the SED Machine (SEDM; Blagorodnova et al. 2018; Rigault et al. 2019; Kim et al. 2022) on the Palomar 60-inch (1.5 m) telescope (P60; Cenko et al. 2006). For AT2024aehp, we also have ultraviolet (UV) data from the *Neil Gehrels Swift Observatory* (*Swift*; Gehrels et al. 2004) Ultraviolet/Optical Telescope (Roming et al. 2005). For AT2024qfm, we also present photometry from the Alhambra Faint Object Spectrograph and Camera (ALFOSC) on the 2.56 m Nordic Optical Telescope (NOT; Djupvik & Andersen 2010), the Inamori-Magellan Areal Camera & Spectrograph (IMACS; Dressler 2010) mounted on the 6.5 m Magellan-Baade telescope, the Large Monolithic Imager (LMI) on the 4.3 m Lowell Discovery Telescope (LDT; Bida et al. 2014), and the Goodman High Throughput Spectrograph (GHTS; Clemens et al. 2004) camera on the 4.1 m Southern Astrophysical Research (SOAR) Telescope[31]. **We correct for Milky Way extinction using the `extinction` Python package[32], choosing the prescription of Schlafly & Finkbeiner (2011) and $R_V = 3.1$.**

The optical light curves of each object are shown in Figure 1. For each object, $t_{\rm rest} = t_{\rm obs} = 0$ was chosen to match the peak of the transient's optical light curve. To convert to absolute magnitude, we apply the conversion

$$M = m_{\rm obs} - 5\log_{10}\left(\frac{D_L}{10\,\mathrm{pc}}\right) + 2.5\log_{10}(1+z)\,. \quad (1)$$

### 2.2. *Optical Spectroscopy*

As part of our follow-up campaigns, we collected optical spectra at $t_{\rm obs} = 4$–37 days. The dates and instruments of all spectroscopic observations are given in Table 3 in Appendix B. We utilized the following facilities: the Gemini Multi-Object Spectrograph (GMOS; Hook et al. 2004, Gimeno et al. 2016), the Keck Low-Resolution Imaging Spectrometer (LRIS; Oke et al.

[31] SOAR2024B-021; PI I. Andreoni
[32] https://github.com/sncosmo/extinction

**Table 1.** LFBOT Summary

| TNS name | Survey name | Redshift | RA (J2000) | Dec (J2000) | Offset (kpc) | $t_0$ (UTC) | Ref. |
|---|---|---|---|---|---|---|---|
| AT2022abfc | ZTF22abvrxkk | 0.212 | 4:51:19.200 | −26:58:41.59 | 2.0 | 2022-11-21 08:15:48 | This work |
| AT2023fhn | ZTF23aaeozpp/ATLAS23hnk | 0.24 | 10:08:03.805 | 21:04:26.70 | 17 | 2023-04-12 05:22:39 | [1,2]; This work |
| AT2023hkw | ZTF23aaimsja | 0.339 | 10:42:17.748 | 52:29:19.03 | 15 | 2023-04-30 04:08:52 | This work |
| AT2023vth | ZTF23ableqsp | 0.0747 | 17:56:34.403 | 8:02:37.30 | 3 | 2023-10-20 03:08:39 | This work |
| AT2024qfm | ZTF24aaxhxhf/PS24hhm | 0.2270 | 23:21:23.450 | +11:56:31.99 | 3.7 | 2024-07-27 08:18:46 | This work; [14] |
| AT2024aehp | ZTF24abygbss | 0.1715 | 8:21:07.474 | 28:44:22.17 | 0.47 | 2024-12-19 10:02:49 | This work |
| — | CSS161010 | 0.033 | 04:58:34.396 | −08:18:03.95 | 0.28 | 2016-10-10 11:31:12 | [3,4] |
| AT2018lug | ZTF18abvkwla | 0.2714 | 02:00:15.194 | +16:47:57.32 | 1.9 | 2018-09-12 09:46:48 | [5] |
| AT2018cow | ATLAS18qqn/ZTF18abcfcoo/Gaia18bqa | 0.014145 | 16:16:00.220 | +22:16:04.91 | 1.7 | 2018-06-16 10:35:02 | [6] |
| AT2020mrf | ZTF20abfhyil/ATLAS20paa | 0.1353 | 15:47:54.163 | +44:39:07.41 | 1.19 | 2020-06-15 08:55:40 | [7] |
| AT2020xnd | ZTF20acigmel | 0.2433 | 22:20:02.030 | −02:50:25.30 | 0.8 | 2020-10-12 04:06:03 | [8] |
| AT2022tsd | ZTF22abftjko/PS22jvi | 0.2564 | 03:20:10.863 | +08:44:55.63 | 6 | 2022-09-07 11:21:22 | [9] |
| AT2024wpp | **GOTO24gjk/ZTF24abjjpbo/ PS24jhj/ATLAS24ond** | 0.0868 | 02:42:05.499 | −16:57:22.90 | 5 | 2024-09-26 01:24:35 | [10–13] |

Note—Basic information on each LFBOT, such as position and redshift. The six LFBOTs discussed in this paper are above the line, while the rest of the known LFBOT population is shown below. $t_0$ is chosen to be the peak of the optical light curve. Positions of the six LFBOTs presented in the paper and AT2020xnd were determined using the methodology in Section 4.1.

**References**—[1] Chrimes et al. (2024a), [2] Chrimes et al. (2024b), [3] Coppejans et al. (2020), [4] Gutiérrez et al. (2024), [5] Ho et al. (2020a), [6] Prentice et al. (2018), [7] Yao et al. (2022), [8] Perley et al. (2021), [9] Ho et al. (2023c), [10] Pursiainen et al. (2025), [11] Nayana et al. (2025), [12] LeBaron et al. (2025), [13] Perley et al. (2026), [14] Nayana A.J. et al. (in prep)

1995) on the 10 m Keck-I telescope, the Keck Cosmic Web Imager (KCWI; Morrissey et al. 2018) and the Deep Imaging Multi-Object Spectrograph (DEIMOS; Faber et al. 2003) on the 10 m Keck-II telescope, the Double Spectrograph on the Palomar 5 m telescope (DBSP; Oke & Gunn 1982), and the Binospec instrument (Fabricant et al. 2019) on the 6.5 m Multiple Mirror Telescope (MMT). Some of these spectra **contain contributions from both sources as they** were taken at a time when the transient and the host galaxy's brightness are comparable **or were created from apertures containing both sources. In particular, the O II, O III, N II, and S II lines are well-known galactic emission lines and indicate the that there is a significant host-galaxy contribution to the spectrum.**

**The basic details for each spectroscopic observation, including the instrument name, observer-frame time after optical peak, PI, and reduction software used, are descripted in Table 3 within Appendix B.** Spectra are shown in Figure 2. Narrow emission lines are observed from the LFBOT host galaxies, (e.g., Ca II H&K, [N II], [O II], [O III], and [S II]). . These spectra also have a contribution from the LFBOT itself, expected to be a featureless blue continuum closely approximated by a blackbody (e.g., Margutti et al. 2019; Perley et al. 2021). The Balmer H$\alpha$ line is prominent in all six hosts.

### 2.3. *Swift X-ray Observations*

X-ray observations were obtained with the *Swift* X-ray Telescope (XRT; Burrows et al. 2005) by ToO triggers shortly after discovery of each LFBOT. Count rates for the XRT instrument were obtained from the UKSSDC *Swift* light-curve generator[33]. Utilizing tools from NASA HEASARC, we converted these count rates to an X-ray flux accounting for the H I column density in the Milky Way. For upper limits (all objects except AT2024qfm), we assume a power-law photon index of $\Gamma = 2$, a typical value for previous LFBOTs (Rivera Sandoval et al. 2018; Margutti et al. 2019; Bright et al. 2022; Matthews et al. 2023).

For AT2024qfm, *Swift* obtained 12 observations, from $t_{\rm obs} \sim 8$–133 d after the first optical detection. We visually inspected the images and discarded one observation (ID 00016748006) because the source landed on the XRT detector's hot column. The X-ray light curve shows a general decline, with the exception of one observation (ID 00016748007) which showed a significant rebright-

[33] UKSSDC *Swift* Light-Curve Generator

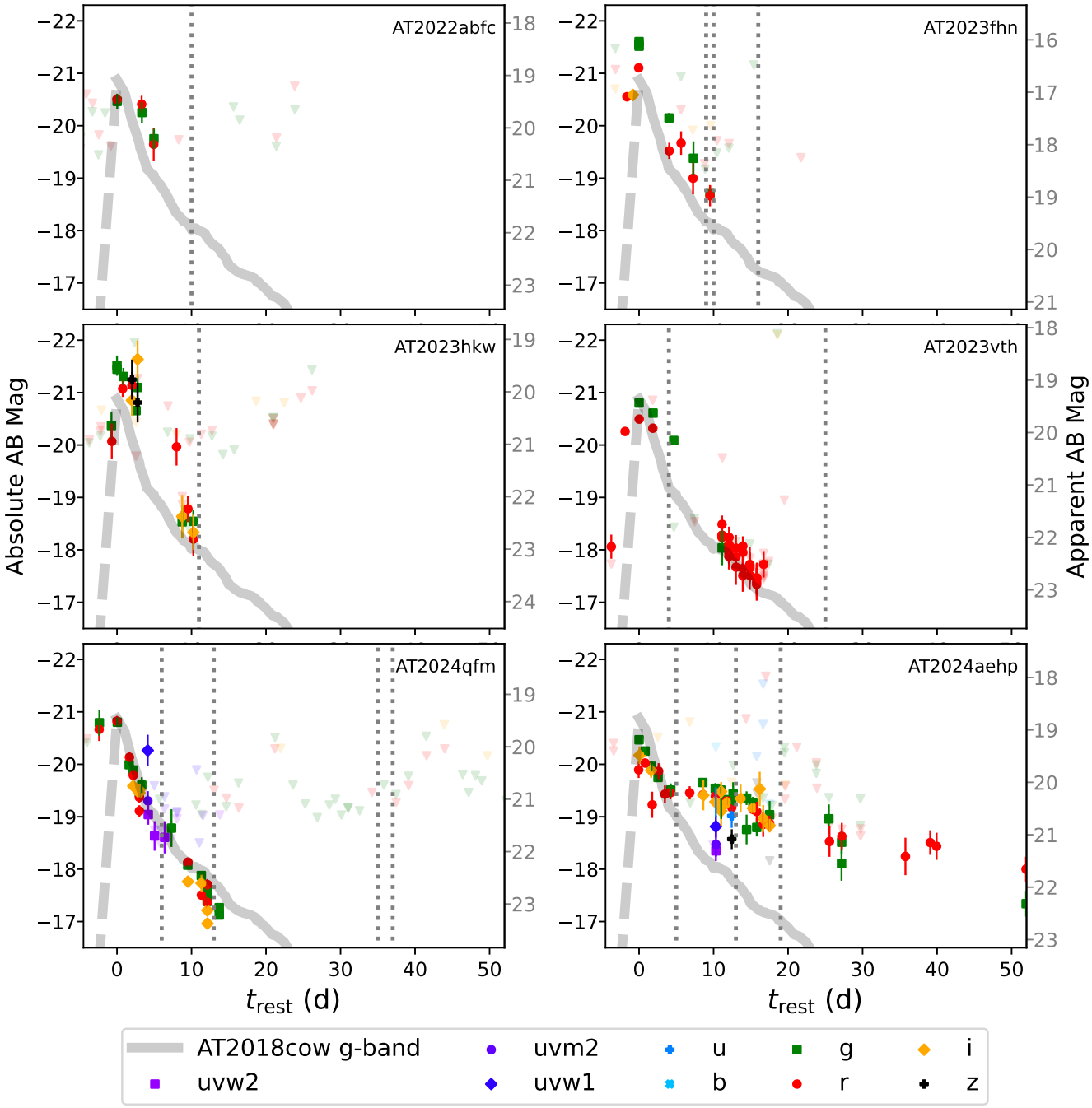


**Figure 1.** Optical light curves of each LFBOT (points), as well as the light curve of AT2018cow (thick gray line, where dashed lines mark the connection to the last upper limit). The left-hand ordinate values display the $E(B-V)$ corrected absolute AB magnitude (Oke & Gunn 1983) and are shared for all plots. The right-hand ordinate displays the apparent AB magnitude. Each filter has a unique marker color and style. Upside-down triangles represent 3-$\sigma$ upper limits. The vertical **dotted** lines show when spectra of the transient and/or its host galaxy were taken. These spectra are displayed in Figure 2.

ening. This observation was extremely short, with 233 s of raw exposure but only 40 s of clean data. Two source counts were detected at the position of AT2024qfm; using the Kraft et al. (1991) approach of the *Swift* tools, the implied detection significance is 99.83% (just over 3-$\sigma$). Closer inspection of the data revealed no clear reason why this would be an erroneous measurement.

For purpose of comparison and for completeness, we also analyze three early *Swift* XRT observations of the LFBOT CSS161010. Late-time (100 d) observations were presented in Coppejans et al. (2020), while up-

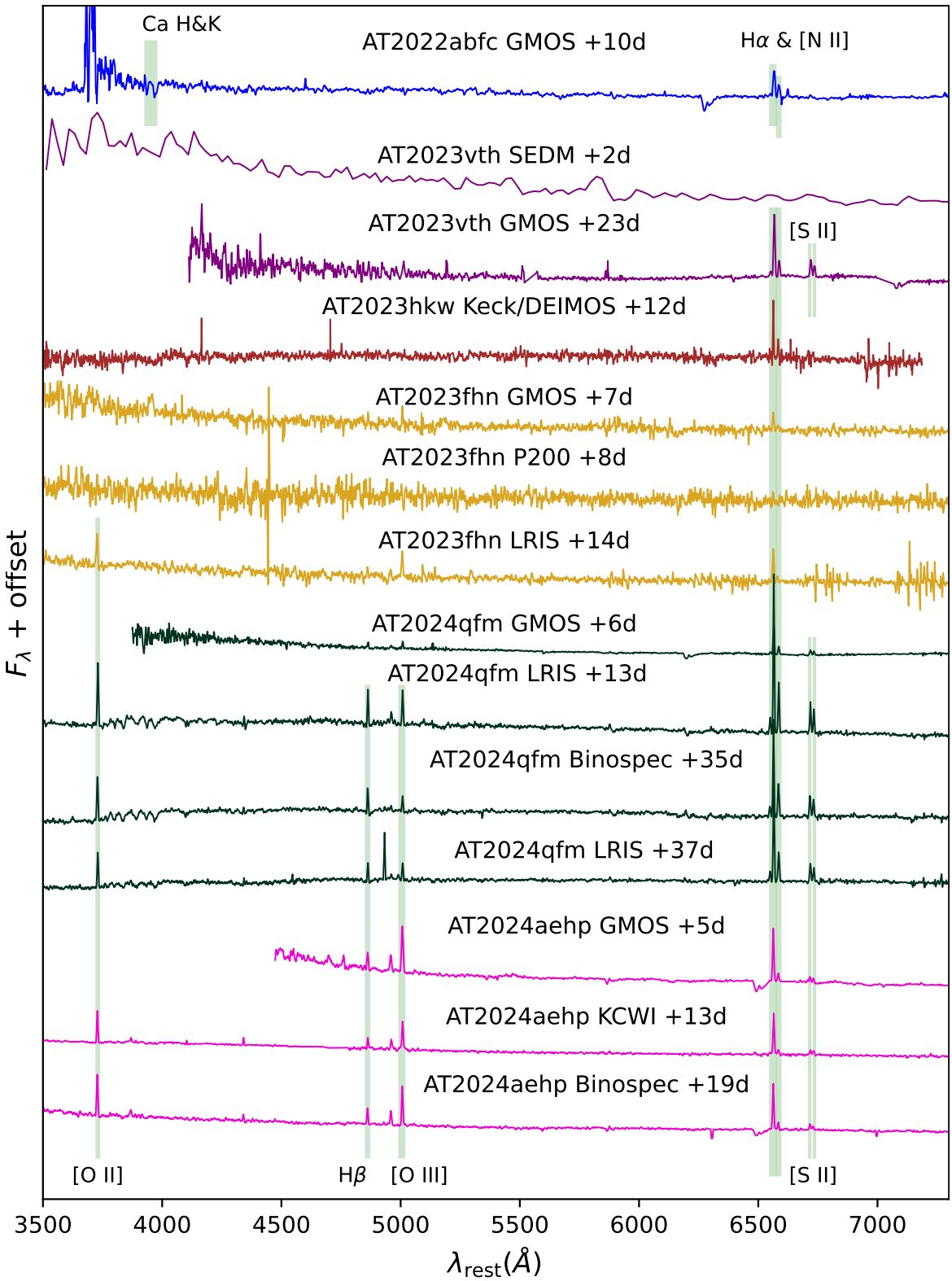


**Figure 2.** Spectra of each LFBOT and/or its host galaxy. For each spectrum we indicate the observer-frame epoch and instrument used. **We bin all spectra with a bin size of 3 Å**. The green shaded rectangles show galactic narrow emission lines that were used to determine a redshift. The spectra are displayed with an arbitrary vertical offset.

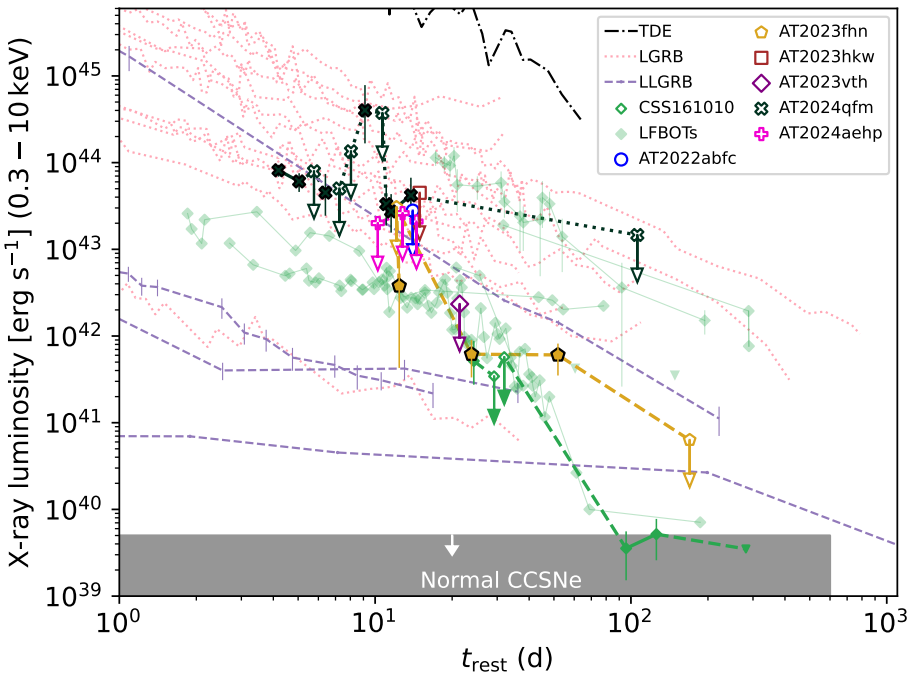


**Figure 3.** 0.3–10 keV X-ray light curves of the six LFBOTs in this paper, in comparison to those of other LFBOTs, **long gamma-ray bursts (LGRBs), and low-luminosity gamma-ray bursts (LLGRBs)**. Open symbols with downward-facing triangles mark 3-$\sigma$ upper limits. Data are from Pian et al. (2000), Tiengo et al. (2004), Campana et al. (2006), Soderberg et al. (2006b), Blanchard et al. (2017), Ho et al. (2020b), Coppejans et al. (2020), Hinkle et al. (2021), Perley et al. (2021),Yao et al. (2022), Ho et al. (2022b), Perley (2023), Ho et al. (2023c), Chrimes et al. (2024a), Yao et al. (2024), and Perley et al. (2026), as well as public *Swift* data.

per limits for these early observations were presented in Gutiérrez et al. (2024) (see their Appendix A.1). We find a very marginal (3.3-$\sigma$) detection in the first epoch, and non detections in the subsequent two epochs. Our upper limits are within a factor **of three of those in Gutiérrez et al. (2024), which can be attributed to the fact that they bin the observations**. We also show the *Chandra* detections of AT2023fhn published by Chrimes et al. (2024b), using the revised values in Nayana et al. (2025). ***Chandra measures X-ray flux over a very similar energy range as* Swift (0.2–10 keV vs. 0.3–10 keV), so these measurements should be comparable.**

The *Swift* XRT observations are summarized in Table 4 in Appendix B. Five of the **new** LFBOTs had a *Swift* non-detection, resulting in upper limits for the photon count rate and the X-ray flux. These upper limits are compared to other objects with *Swift* detections in Figure 3.

### 2.4. *Radio and Millimeter Observations*

We obtained radio observations with the National Science Foundation's Karl G. Jansky Very Large Array (VLA; Thompson et al. 1980; Perley et al. 2011), the Giant Metrewave Radio Telescope (GMRT; Swarup et al. 1991), and the Northern Extended Millimeter Array (NOEMA). All of these observations are listed in Table 5 in Appendix B.

#### 2.4.1. *VLA*

We present VLA observations for five LFBOTs[34], all of which were detected. Radio data for AT2024qfm will be published by Nayana A.J. et al. in prep **as per a prior arrangment**. Observations were obtained at mid-frequencies of 3 GHz (S-band), 6 GHz (C-band), 10 GHz (X-band), 16 GHz (Ku-band), and 33 GHz (Ka-band). Based on the behavior of past sources **(Yao et al. 2022; Ho et al. 2022c; Chrimes et al. 2024a)**, 10 GHz (X-band) gives the most reliable chances of detection, so every epoch features an observation in this band. We calibrate VLA data using the automated pipeline in Common Astronomy Software Applications package version 6.5.4-9 (CASA; Team et al. 2022). We use Briggs weighting with `robust = 0.5` (Briggs 1995).

[34] AT2022abfc was observed under programs 22A-405 (PI D. Perley), 23A-393, 23A-422, and 23B-138; AT2023fhn under programs 23A-403 and 23B-138; AT2023vth under program 23B-138; AT2023hkw under program 23A-415; AT2024aehp under programs 24B-338 and 25A-367 (PI A. Ho)

When the image quality is degraded by bright artifacts from nearby luminous sources, we perform phase-only self-calibration on these other sources. Flux measurements were determined by using the Cube Analysis and Rendering Tool (CARTA; Comrie et al. 2021). For detections, the flux density is the maximum intensity at the transient's position, and the uncertainty is the measured root-mean square (RMS) in the background. For non-detections, we take the upper limit to be 3 × RMS. The radio light curves are shown in Figure 4 and the radio SED evolution is shown in Figure 5.

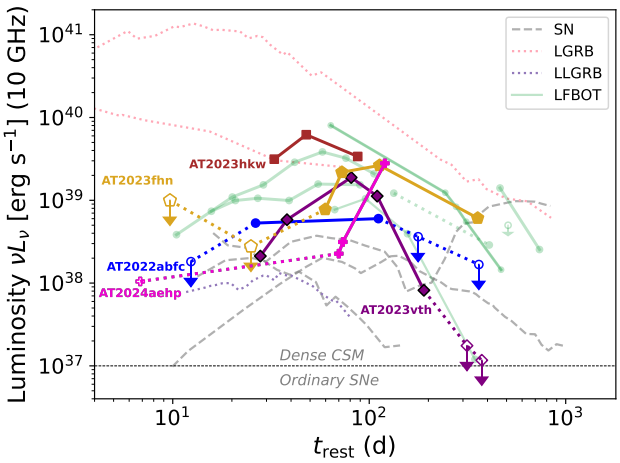


**Figure 4.** Radio light curves at 10 GHz of the LFBOTs with data presented in this paper, compared with other known LFBOTs, long GRBs (LGRBs), low-luminosity GRBs (LLGRBs), and core-collapse supernovae with non-relativistic ejecta (SNe). We measure time after peak light in the transient's rest frame. Data from Kulkarni et al. (1998), Berger et al. (2003), Soderberg et al. (2006a), van der Horst et al. (2008), Soderberg et al. (2010b), Salas et al. (2013), Margutti et al. (2019), Coppejans et al. (2020), Ho et al. (2022c), Yao et al. (2022), Perley et al. (2021), Ho et al. (2023c), and Nayana et al. (2025).

### 2.4.2. *GMRT*

We also present GMRT observations of AT2022abfc, AT2023fhn, AT2023hkw, and AT2023vth. Both AT2023fhn and AT2023vth were detected. These observations were spread across three frequency bands: Band-3 (250-500 MHz), Band-4 (550-850 MHz), and Band-5 (1000-1460 MHz)[35]. The GMRT data were reduced through CASA and AIPS. We use Briggs weighting with `robust = 0.5`. The primary beam correction was done following the method described in the CASA documentation[36]. We measure the flux density using the `imfit` method in the CASA documentation, which provides the maximum intensity for a fitted Gaussian. If the measured flux is not 3-$\sigma$ above the background flux RMS, we record an upper limit of 3× the RMS.

### 2.4.3. *NOEMA*

We observed five LFBOTs using NOEMA[37] (all but AT2023hkw), at times ranging from $t$ = 10–100 d, resulting in detections of three objects (AT2024aehp, AT2024qfm, AT2023vth) and non-detections of two objects (AT2022abfc, AT2023fhn). The data reduction was done with Continuum and Line Interferometer Calibration (`CLIC`) and `MAPPING`, which are part of the `GILDAS` software package (Gildas Team 2013)[38]. Unless otherwise specified, all uncertainties are 1$\sigma$, and **the systematic error of the** flux scale is 10% in the 3 mm band and 15% at 1.3 mm. The NOEMA data are provided in Table 5 and displayed in Figure 6.

AT2024aehp was observed on May 23, 2025 (> 100 d after optical discovery; we triggered mm observations following the cm-wave detection) in the 10D configuration under mediocre weather conditions, with MWC349 as the flux calibrator. Reduction was performed by hand and we estimate a 15% systematic calibration uncertainty for the absolute flux scale.

AT2023fhn was observed on April 22, 2023 in the 12C configuration under good conditions, with LKHA101 as the flux calibrator. The image included a nearby bright source, and subtracting the contribution from the bright source did not result in a detection of AT2023fhn at the phase center.

An observation of AT2022abfc was triggered on December 30, 2022; an attempt was made on January 7, 2023, but could not be carried out owing to bad weather. A successful observation was obtained on January 10 in the 12A configuration, with LKAH101 as the flux calibrator. The source was not detected, but limits were weak because the source was at high airmass and the array was in an extended configuration.

AT2023vth was first observed on November 20, 2023 in the 9C configuration, under cloudy conditions, with the flux calibrator MWC349. The source was detected, so we pursued multifrequency follow-up observations.

[35] AT2022abfc was observed under program ID 44_099. AT2023fhn was observed under programs 45_117 and ddtC299. AT2023hkw was observed under 45_117 and ddtC294. AT2023vth was observed under program 45_113 (PI Nayana A.J.).

[36] CASA: Imaging with Custom Primary Beams

[37] Program IDs W22BT (AT2023fhn and AT2022abfc), W24EW (AT2024aehp), S23BF (AT2023vth), D24AA (Director's Discretionary Time; AT2024qfm), all PI A. Ho.

[38] https://www.iram.fr/IRAMFR/GILDAS

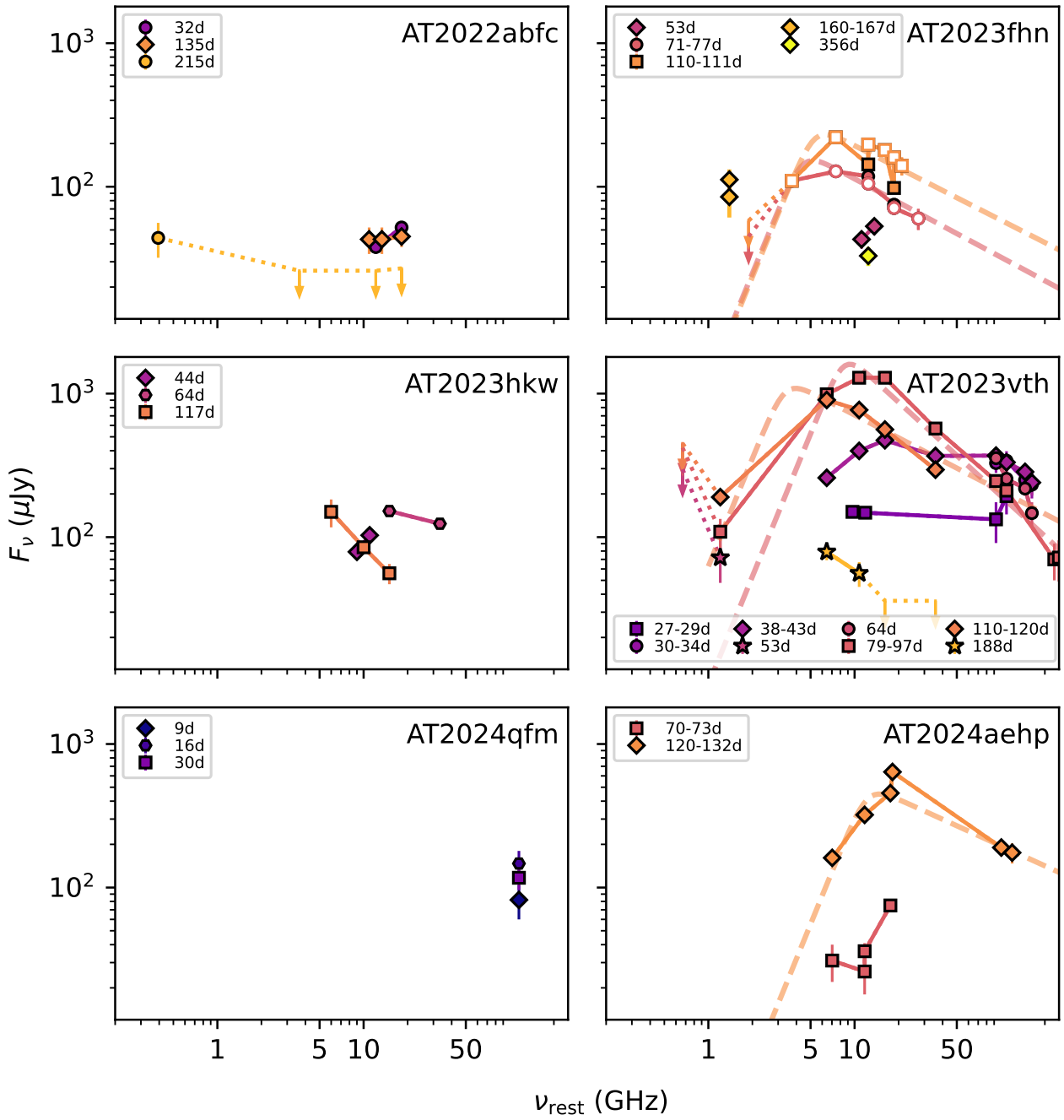


**Figure 5.** The radio and millimeter SED of each LFBOT at different observer-frame epochs. Observations taken within a few days are grouped into one epoch. Non-detections are signified by **a downward pointing arrow**. For clarity, we omit some non-detections. For epochs with sufficient data points, we display the broken power-law fit to the SED as a dashed line. Data from Chrimes et al. (2024a) for AT2023fhn **have markers filled in with white.**

The timing of these observations was affected by weather conditions, particularly at the highest frequencies.

NOEMA observed AT2024qfm on four dates, in compact configuration (D), with primary flux calibrator MWC349. The first observation was on August 5, 2024. The source was detected in the first three epochs.

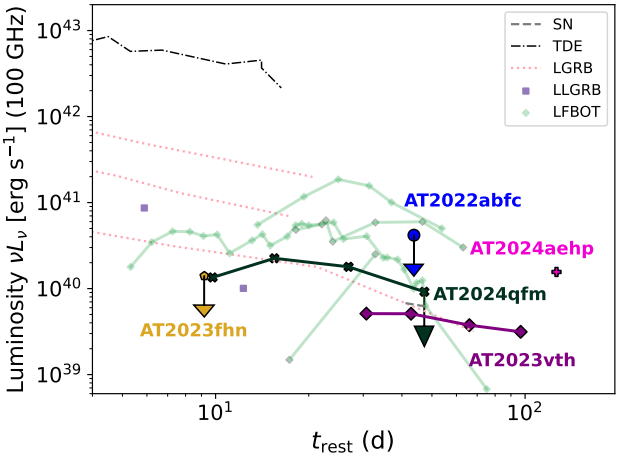


**Figure 6.** Millimeter-wave (100 GHz) light curves in comparison to other LFBOTs, long-duration gamma-ray bursts (LGRBs), low-luminosity GRBs (LLGRBs), and tidal disruption events (TDEs). AT2022abfc and AT2023fhn only have an upper limit. AT2023fhn has been shifted slightly to the left for clarity. Data are from Kulkarni et al. (1998); Zauderer et al. (2011); Perley (2013); Corsi et al. (2014); Laskar et al. (2016); Perley et al. (2017); Laskar et al. (2018); Margutti et al. (2019); Bright et al. (2022); Ho et al. (2023c); Nayana et al. (2025).

### 2.5. *Search for Gamma-Ray Bursts*

We performed an archival search for GRB detections in the NASA General Coordinates Network Circulars[39]. We looked for GRBs detected within a week of the transient's optical light-curve peak **that had a localization that was within 3-$\sigma$ of the transient's coordinates**. No such GRBs were reported.

### 2.6. *Search for X-ray Precursors*

For the model by Metzger (2022) with runaway mass transfer leading up to the dynamical merger of a stripped star and compact-object companion, one might expect X-ray and UV/optical precursor emission before the LFBOT (Tsuna et al. 2024). **This could be a test of the merger model; other models such as core collapse do not make a similar prediction.** We searched for X-ray precursor emission in the eROSITA-DE Data Release 1 (DR1). eROSITA (Predehl et al. 2021) is the soft X-ray instrument onboard the orbital observatory Spectrum-Roentgen-Gamma (SRG; Sunyaev et al. 2021). DR1 comprises data from the first six months, which ended in June 2020 (Merloni et al. 2024; Tubín-Arenas et al. 2024).

Of the thirteen LFBOTs in Table 1, three were in the footprint of eROSITA-DE DR1: AT2022abfc, CSS161010, and AT2024wpp. For each object, we used the products from Tubín-Arenas et al. (2024) and Merloni et al. (2024) to determine if there was a detection at the source position, and if not, obtained a 0.2–2.3 keV upper limit (the most sensitive energy range for eROSITA). **As each field was only imaged once in this data release, we only check for consistency in position and allow the observation to be at any time.** The closest catalogued sources were 5.5′ (AT2022abfc), 3′ (CSS161010), and 2′ (AT2024wpp), too distant to be associated **(the spatial resolution of eROSITA is $< 10''$, Predehl et al. 2021)**.

For AT2022abfc and AT2024wpp (the two objects for which the eROSITA observations were prior to the LFBOT), the limits are $9.1 \times 10^{42}$ erg s$^{-1}$ corresponding to 2.7 yr pre-LFBOT, and $9.0 \times 10^{41}$ erg s$^{-1}$ corresponding to 4.7 yr pre-LFBOT, respectively. For CSS161010, the eROSITA limit is $1.5 \times 10^{41}$ erg s$^{-1}$, 3.4 yr after the LFBOT. The next eROSITA data release (scheduled for mid-2026) will provide more stringent upper limits (i.e., closer to the LFBOT time).

### 2.7. *Host-Galaxy Photometry*

IR and UV photometry were obtained for all LFBOT host galaxies from the Near-Earth Object Wide-field Infrared Survey Explorer mission (NEOWISE; Mainzer et al. 2011) on the *Wide-field Infrared Survey Explorer* space telescope (*WISE*), and the *Galaxy Evolution Explorer* (*GALEX*; Martin et al. 2005), respectively. Optical photometry for the host galaxies was obtained from the Legacy Survey Data Release 10 (Dey et al. 2019). The host galaxy of AT2023vth does not have Legacy Survey optical photometry, so we used optical photometry from Pan-STARRS DR2 (PS1; Chambers et al. 2016). We also used *Hubble Space Telescope (HST)* photometry from Chrimes et al. (2024b) for AT2023fhn. All the *GALEX* data used in this paper can be found in MAST: 10.17909/zktz-mn64. For the photometry in all bands, we required a signal-to-noise ratio (SNR) greater than 5 for a detection. **We choose this SNR to include several marginal GALEX and WISE detec-**

[39] https://gcn.nasa.gov/circulars/

**tions, while excluding a few measurements with errors of over 0.5 mag. Chooser a stricter cutoff would the fit far more uncertain as we lose a significant number of data points, but a looser cutoff would have little effect as the new points would be highly uncertain, which the fitting process accounts for. We display cutouts of the host galaxies in Figure 7.**

We used the Python package `gPhoton` (Million et al. 2016) to retrieve *GALEX* near-UV and far-UV photometry for the LFBOT host galaxies. In `gPhoton`, we chose the observation with the longest exposure time and recorded the background-corrected magnitudes. We chose an aperture radius of 6.12 arcseconds for the source, and measured the background from an annulus with an inner radius of 10.8 arcseconds and an outer radius of 21.6 arcseconds. **We estimate that this size would exclude all of the host galaxy's emission and include as much background as possible for an accurate background flux estimate without including any flux from other sources. We apply this annulus size to most of the LFBOT hosts to keeps things as consistent as possible.** In the cases of AT2023hkw and AT2023vth, there were **nearby UV-bright sources that required us to use a smaller annulus**. The inner and outer annulus radii are 10.8 and 16.2 arcseconds for AT2023hkw, and 7.2 and 10.8 arcseconds for AT2023vth. Legacy Survey and *NEOWISE* photometry was obtained with the Python package `noaodatalab`[40] version 2.20.0. Pan-STARRS DR2 *grizy* photometry was obtained from the online catalog search tool[41].

The host-galaxy photometry is provided in Table 6 in Appendix B.

## 3. RADIO ANALYSIS

### 3.1. *Radio Synchrotron Modeling*

We model the SED at radio and millimeter wavelengths as the result of synchrotron emission. Synchrotron radiation with a self-absorbed component forms a broken power-law radio SED. We follow the prescription of Chevalier (1998) to derive properties of the shock and the medium it traverses, such as the shock radius, ambient magnetic field, average shock velocity, pre-shock density, and total shock energy. This is the same procedure used in modeling the radio SEDs for other AT2018cow-like FBOTs (e.g., Margutti et al. 2019; Ho et al. 2019; Chrimes et al. 2024b; Nayana et al. 2025; Perley et al. 2026). We note that in some of these works, some equations are given in terms of angular diameter distance $D_\theta$, but should be given in terms of luminosity distance ($D_L$; private communication) as in the expressions below. **We also note that the following prescription assumes a nonrelativistic shock and does not account for relativistic effects (see, e.g., Margalit & Quataert 2024; Ferguson & Margalit 2025), which become relevant for $\Gamma\beta \sim 1$. For self-consistency, we estimate the $\Gamma\beta$ this model returns and we find that some objects do approach the trans-relativistic regime Fig 8, so we caution against taking our presented results at face-value. Still, we use this method for comparison with previous LFBOTs, which is the main goal of this paper's analysis**.

As in Soderberg et al. (2010a), we assume equipartition of energy between protons, electrons, and the magnetic field; thus, the energy densities in electrons ($\epsilon_e$) and magnetic fields ($\epsilon_B$) are given by $\epsilon_e = \epsilon_B = 1/3$. If we model the source of the emission as a disk with radius $R$ and thickness $S$, then its total volume is $\pi R^2 S$. We then parameterize this as equivalent to a spherical volume $4\pi f R^3/3$, where $f$ is the filling factor. We take $f = 0.5$ as in previous LFBOT studies (Yao et al. 2022; Bright et al. 2022), but the dependences of the equations on this parameter is weak. We assume an electron power-law index of 3, following Chevalier & Fransson (2006); this is a typical value for previous LFBOTs **as determined from fits to the slope of the self-absorbed end of the synchrotron SED (Coppejans et al. 2020; Ho et al. 2019). Through most of our SEDs, we do not have more than two points in the self-absorbed regime, so we use this fiducial value with the recognition that the results do not strongly depend to the power-law index (Chevalier 1998).**

Finally, we verify that the self-absorption frequency is less than the cooling frequency, which is defined as the frequency where the electrons have lost an amount of energy equal to their total energy through synchrotron cooling. We use Equation (C13) of Ho et al. (2022c), which gives the ratio between the self-absorption and cooling frequencies, and confirm that the self-absorption frequency is lower for all cases.

Under these assumptions, Equations (13) and (14) from Chevalier (1998) reduce to these expressions for

[40] https://github.com/astro-datalab/datalab
[41] https://catalogs.mast.stsci.edu/panstarrs/

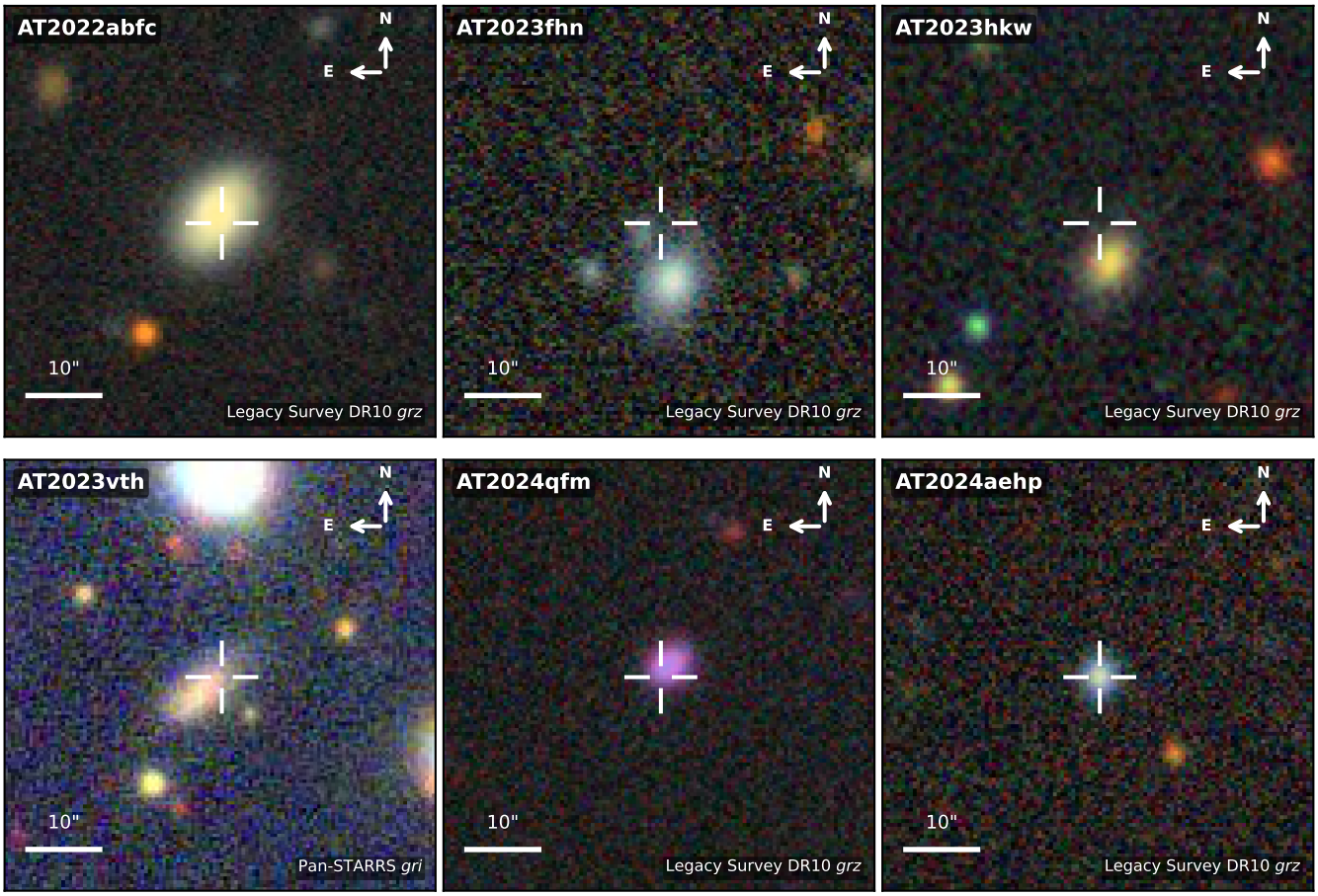


**Figure 7. Optical cutouts of the six LFBOT host galaxies we discuss in this paper.**

the shock radius and magnetic field:

$$R_p = 8.8 \times 10^{15} \left(\frac{\epsilon_e}{\epsilon_B}\right)^{-1/19} \left(\frac{f}{0.5}\right)^{-1/19} \left(\frac{F_\mathrm{p}}{(1+z)\,\mathrm{Jy}}\right)^{9/19} \times \left(\frac{D_L}{\mathrm{Mpc}}\right)^{18/19} \left(\frac{\nu_\mathrm{p}}{5\,\mathrm{GHz}}\right)^{-1} \mathrm{cm}\,, \tag{2}$$

$$B = 0.58 \left(\frac{\epsilon_e}{\epsilon_B}\right)^{-4/19} \left(\frac{f}{0.5}\right)^{-4/19} \left(\frac{F_\mathrm{p}}{(1+z)\,\mathrm{Jy}}\right)^{-2/19} \times \left(\frac{D_L}{\mathrm{Mpc}}\right)^{-4/19} \left(\frac{\nu_\mathrm{p}}{5\,\mathrm{GHz}}\right) \mathrm{G}\,, \tag{3}$$

where $D_L$ is the luminosity distance, $\nu_\mathrm{p}$ **and** $F_\mathrm{p}$ **are the peak frequency and flux density at peak frequency of the radio SED in the transient's rest frame, respectively.**

Following Ho et al. (2019), this can then be combined into an equation for total energy $U = U_B/\epsilon_B$:

$$U = \frac{B^2}{8\pi}\frac{4\pi f R^3}{3}\frac{1}{\epsilon_B} = 1.9 \times 10^{46} \left(\frac{1}{\epsilon_B}\right) \left(\frac{\epsilon_e}{\epsilon_B}\right)^{-11/19} \left(\frac{f}{0.5}\right)^{8/19} \times \left(\frac{F_\mathrm{p}}{(1+z)\,\mathrm{Jy}}\right)^{23/19} \left(\frac{D_L}{\mathrm{Mpc}}\right)^{46/19} \left(\frac{\nu_\mathrm{p}}{5\,\mathrm{GHz}}\right)^{-1} \mathrm{erg}\,. \tag{4}$$

We estimate the average velocity as $R/(t_\mathrm{obs}/(1+z)) = R/t_\mathrm{rest} = \Gamma\beta c$, where $\beta = v/c$ and $\Gamma$ is the Lorentz factor.

Following Chrimes et al. (2024b), we derive formulas for the CSM electron number density $n_e$ and the wind density $\dot{M}/v_w$. This assumes the CSM resulted from a spherical stellar wind with mass-loss rate $\dot{M}$ and wind velocity $v_w$, where $\dot{M}/v_w = 4\pi m_p R^2 n_e$ and $m_p$ is the proton mass. Thus,

$$n_e = 1.02 \frac{1}{\epsilon_B} \left(\frac{\epsilon_e}{\epsilon_B}\right)^{-6/19} \left(\frac{f}{0.5}\right)^{-6/19} \left(\frac{F_\mathrm{p}}{(1+z)\,\mathrm{Jy}}\right)^{-22/19} \times \left(\frac{D_L}{\mathrm{Mpc}}\right)^{-44/19} \left(\frac{\nu_\mathrm{p}}{5\,\mathrm{GHz}}\right)^{4} \left(\frac{t_\mathrm{rest}}{\mathrm{days}}\right)^{2} \mathrm{cm}^{-3}, \tag{5}$$

$$\frac{\dot{M}}{v_w} \left(\frac{1000\,\mathrm{km\,s^{-1}}}{10^{-4}\,M_\odot\,\mathrm{yr}^{-1}}\right) = 2.5 \times 10^{-5} \left(\frac{1}{\epsilon_B}\right) \left(\frac{\epsilon_e}{\epsilon_B}\right)^{-8/19} \times \left(\frac{f}{0.5}\right)^{-8/19} \left(\frac{F_\mathrm{p}}{(1+z)\,\mathrm{Jy}}\right)^{-4/19} \times \left(\frac{D_L}{\mathrm{Mpc}}\right)^{-8/19} \left(\frac{\nu_\mathrm{p}}{5\,\mathrm{GHz}}\right)^{2} \times \left(\frac{t_\mathrm{rest}}{\mathrm{days}}\right)^{2}. \tag{6}$$

The peak frequency and flux density of the SED at a given epoch are measured by fitting a smoothed broken-power law of the form

$$F(\nu) = F_\mathrm{p} \left((\nu/\nu_\mathrm{p})^{-sa_1} + (\nu/\nu_\mathrm{p})^{-sa_2}\right)^{-1/s}. \tag{7}$$

We choose $s = 1$ for the smoothness parameter, as was done in previous LFBOT studies (Bright et al. 2022; Chrimes et al. 2024a; Nayana et al. 2025). We also set $a_1 = 5/2$ to fix the power-law slope of the self-absorbed half of the SED, as required by Chevalier's formulation. When an epoch involves observations from multiple observations, **we apply an extra error equal to 10% of the flux density in quadrature to account for systematic effects between arrays that are unaccounted for**. A fit is performed only when we have detections on either side of the SED's peak. The best-fit parameters from the broken power-law fit are provided in Table 7 in Appendix B. If we are unable to perform a broken power-law fit, we use the values at the detection closest to the peak's implied location in place of the frequency and flux density at peak, and calculations based on these values will instead set limits on the synchrotron parameters. The inferred synchrotron parameters are listed in Table 8 in Appendix B, and the results are displayed in Figure 8. This includes epochs where we do fit a broken-power law and have point estimates for parameters, and epochs where we can only constrain the peak's location and place limits on parameters instead. We find that the implied CSM densities for these six LFBOTs are comparable to those derived in the literature for previous LFBOTs, with values of 10–100 $\mathrm{cm}^{-3}$ at radii of $10^{17}$cm.

## 4. HOST-GALAXY ANALYSIS

### 4.1. *Host-Galaxy Offsets*

**With an expanded sample of 13 LFBOTs, we also examine the distribution of LFBOT offsets from the host-galaxy nuclei. We choose the host galaxy as the galaxy closest in offset to the transient, and then confirm that there are no lines at conflicting redshifts in spectroscopy of the LFBOT and its host.** For AT2023fhn, we used the localization from *HST* imaging presented by Chrimes et al. (2024b). For AT2024qfm, we use the localization from the ZTF data as we do not present radio data for that object in this paper. For the other objects, including AT2020xnd (which did not have a published offset) we obtained the transient's position from a Gaussian fit in the image of the LFBOT's brightest radio detection. We measure the host galaxy's position from Legacy Survey Data Release 10, except for AT2023vth's host which was measured from Pan-STARRS Data Release 2. These offsets are recorded in Table 1.

In Figure 9, we display the cumulative distribution of all LFBOT offsets in kpc compared with other classes of extragalactic explosive transients. The offset distribution appears to be intermediate to that of CCSNe and that of LGRBs and SLSNe. **While the LFBOT sample size is still small, it is almost double that of previous analyses, giving us greater confidence in our results**.

### 4.2. *Host-Galaxy Properties*

We used the Python package `prospector` (Leja et al. 2017; Johnson et al. 2021)[42] to derive properties such as stellar mass and star-formation rate (SFR) from the host-galaxy photometry. The `prospector` package utilizes the Flexible Stellar Population Synthesis[43] (FSPS) package to model the emission from each galaxy given a set of parameters for the galaxy's composition and stellar population. **It also utilizes dynamic nested sampling through the package `dynesty`[44] to fit galaxy parameters to the observed galaxy photometry under Bayesian inference.** The basic parametric star-formation history template is adopted along with the parameters for nebular emission. We chose the Chabrier initial mass function (Chabrier 2003), the Calzetti dust attenuation model (Calzetti et al. 2000), and a star-formation history of the form $te^{-t/\tau}$. **These**

[42] https://github.com/bd-j/prospector
[43] https://dfm.io/python-fsps/current/
[44] https://github.com/joshspeagle/dynesty

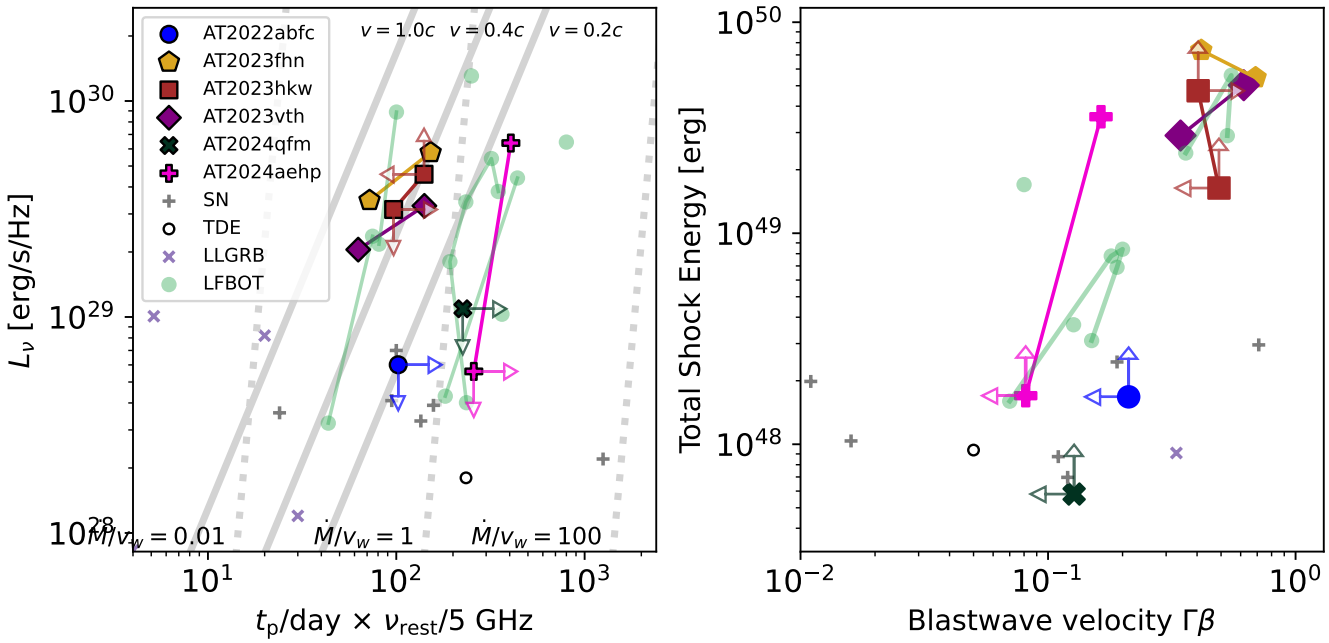


**Figure 8.** Inferred parameters from our synchrotron self-absorption modeling. Some epochs do not include the self-absorption peak and merely constrain its location; the corresponding upper limits on parameters are marked with arrows. Values for other objects are obtained from Ho et al. (2019) and Ho et al. (2023c); and Perley et al. (2026) for AT2024wpp. We label each object and the observer-frame time after peak light. *Left:* Luminosity density at $\nu_p$ versus peak frequency and time after explosion. We also display lines of constant shock velocity and $\dot{M}/v_w$, where $v_w = 1000\,\mathrm{km\,s^{-2}}$ and $\dot{M}$ is in units of $10^{-4}\,M_\odot\,\mathrm{yr}^{-1}$. *Right*: Total shock energy derived from Equation 4 against shock velocity in terms of $\Gamma\beta$.

**Table 2.** Host-Galaxy Properties

| Object | Mass ($\log M/M_\odot$) | Metallicity ($\log Z/Z_\odot$) | Age (Gyr) | SFR ($M_\odot\,\mathrm{yr}^{-1}$) | Host Extinction ($\tau$) |
|---|---|---|---|---|---|
| AT2022abfc | $10.80^{+0.02}_{-0.04}$ | 0.05 | $1.0^{+0.1}_{-0.2}$ | $4.3^{+0.6}_{-0.6}$ | $0.83^{+0.04}_{-0.04}$ |
| AT2023fhn | $10.07^{+0.06}_{-0.06}$ | −0.27 | $0.7^{+0.5}_{-0.2}$ | $7.7^{+1.4}_{-1.1}$ | $0.59^{+0.05}_{-0.05}$ |
| AT2023hkw | $10.7^{+0.1}_{-0.1}$ | 0.03 | $2.7^{+1.2}_{-0.8}$ | $2.8^{+1.3}_{-0.8}$ | $0.22^{+0.17}_{-0.14}$ |
| AT2023vth | $8.95^{+0.03}_{-0.04}$ | −0.60 | $0.62^{+0.13}_{-0.07}$ | $0.24^{+0.04}_{-0.03}$ | $0.0035^{+0.0055}_{-0.0026}$ |
| AT2024qfm | $10.2^{+0.2}_{-0.1}$ | −0.20 | $4.4^{+5.0}_{-3.1}$ | $3.3^{+1.1}_{-1.1}$ | $0.60^{+0.14}_{-0.13}$ |
| AT2024aehp | $8.9^{+0.1}_{-0.1}$ | −0.60 | $0.8^{+0.3}_{-0.3}$ | $1.6^{+0.2}_{-0.2}$ | $0.63^{+0.05}_{-0.05}$ |

NOTE—Results of our `prospector` host-galaxy fits. Metallicities were not determined by the fitting process, but by using a preliminary run at $\log Z/Z_\odot = -0.2$, and then converting the mass estimate to a metallicity using Gallazzi et al. (2005). For the host galaxy extinction, we report the `dust2` parameter. FSPS defines this parameter as the optical depth $\tau$ in an extinction factor of $e^{-\tau}$ at the wavelength of 5500 Å.

**procedures, were chosen to match LFBOT host galaxy analysis in previous papers as closely as possible (Yao et al. 2022; Ho et al. 2023c; Chrimes et al. 2024a).** A model was first run with a fixed metallicity of $\log(Z/Z_\odot) = -0.2$. Then, we used the median mass and the mass-metallicity relation of Gallazzi et al. (2005) to fix the metallicity to a more accurate value, and ran a final model. The fitted parameters are listed in Table 6, the best-fit photometry and spectroscopy are displayed in Figure 10, and the fitted host-galaxy properties are listed in Table 2. We show the median host-galaxy masses and SFRs in Figure 11. Our host-galaxy properties derived for the host of AT2023fhn agree with Chrimes et al. (2024a) to within 1-$\sigma$ for host-galaxy mass and 2-$\sigma$ for SFR.

The host galaxies of our LFBOTs are all star-forming galaxies. Whereas several previous events were in low-mass dwarf galaxies (Ho et al. 2020a; Coppejans et al.

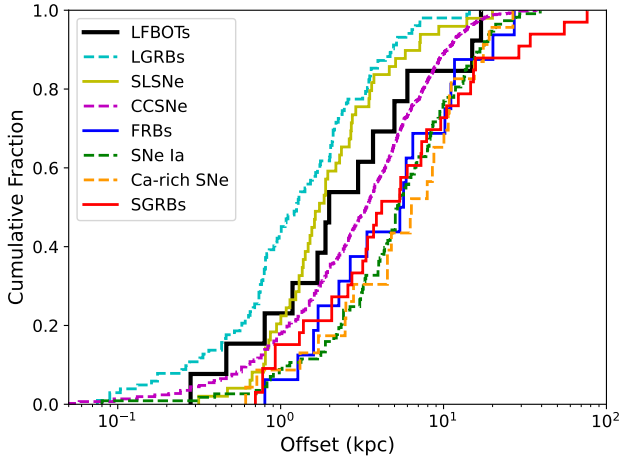


**Figure 9.** The cumulative distribution of host-galaxy offsets for LFBOTs, as well as fast radio bursts (FRBs; Bhandari et al. 2022), LGRBs (Blanchard et al. 2016; Lyman et al. 2017), short GRBs (SGRBs; Fong et al. 2022), CCSNe (Kelly & Kirshner 2012; Schulze et al. 2021), Type Ia SNe (Wang et al. 2013), SLSNe (Lunnan et al. 2015; Schulze et al. 2021), and Ca-rich transients (De et al. 2020). The offsets for previous LFBOTs are from Prentice et al. (2018); Coppejans et al. (2020); Ho et al. (2020a); Yao et al. (2022); Chrimes et al. (2024b).

2020; Chrimes et al. 2024b; Margutti et al. 2019), the host galaxies of four of our new LFBOTs have higher masses ($10^{10}$–$10^{11}$ $M_\odot$, Figure 11). However, our selection criteria for identifying LFBOT candidates do require there to be a visible host galaxy (to avoid contamination by dwarf novae), so a bias to higher mass galaxies **(and therefore higher star-formation rate)** is to be expected.

To compare the *intrinsic* distribution of host-galaxy mass to that of CCSNe, we used the untargeted sample of 150 CCSN host galaxies from Taggart & Perley (2021) as representative of the true CCSN host galaxy population. **We prioritized having an unbiased sample over pure sample size, as our LFBOT sample size is small regardless.** For each of the thirteen LFBOT host galaxies, we calculate the minimum absolute magnitude for it to have been detected by the Legacy Survey at that redshift, which corresponds to the apparent magnitude cutoff of $m_r \leq 23$. Then, for each CCSN host galaxy, we apply a weight equal to the proportion of the thirteen cutoffs that the CCSN host galaxy would pass. We do not account for any systematic differences in host galaxy properties between those at redshift $z \approx 0.01$ as in the Taggart sample, and those at $z \approx 0.1$–$0.2$ as with most of the LFBOTs. We plot the Gaussian kernel density estimation (KDE) for the weighted distribution of CCSN host-galaxy masses Figure 12 as a dotted red line. The Gaussian KDE for the LFBOT host-galaxy masses is shown in black. We repeated the exercise for SLSNe, which are known to have a preference for lower galaxy masses than regular CCSNe (Taggart & Perley 2021).

We use the `Ecume` package (Roux de Bezieux 2024) in the programming language `R` (R Core Team 2025) to perform a Kolmogorov-Smirnov (KS) test comparing the distribution of LFBOT host-galaxy masses with the weighted CCSN and SLSN populations. These simulations imply that the LFBOT host-galaxy population is consistent with both the populations of CCSN and SLSN host galaxies at a significance of $\alpha = 0.05$ (**which means that we required the p-value to be less than** 0.05 **to conclude that the underlying distributions are different at 95% confidence**). **We also display the results of the KS test in Figure 12.**The SLSN host-galaxy distributions appears especially similar to that of LFBOTs. However, the SLSNe are subdivided into SLSNe-I and SLSNe-II. These two classes are markedly different, and this is reflected in their host-galaxy masses—SLSN-I host galaxies are clustered around $\log(M/M_\odot) \approx 8$ while SLSN-II hosts are $\log(M/M_\odot) \approx 9$–$10$. Despite the visible differences, the sample sizes of SLSN-I and SLSN-II hosts are too small to find a statistically significant difference at $\alpha = 0.05$.

## 5. DISCUSSION

### 5.1. *Observational Comparison*

The observations presented in Section 2 show relative homogeneity in the optical (Figure 1) and radio (Figure 4) light curves. To some extent the similarity of the optical light curves to those of AT2018cow is by selection, as these objects were identified on the basis of a rapid rise and decay **within the 2–3 days of the optical peak,** along with high luminosities. The rapid decline of 3–4 mag over $t_{\rm rest} \approx 10$–**15** d observed in AT2023fhn, AT2023vth, AT2023hkw, and AT2024qfm — very similar to AT2018cow — was not selected for. **We note that AT2024aehp differs in many of these properties. This section will focus on the other LFBOTs in our sample; AT2024aehp will be discussed in more detail in Section 5.3.**

The similarity between the radio light curves is less of a selection effect. Figure 4 shows that the peak $\nu L_\nu$ values mostly lie within the range of $10^{39}$–$10^{40}$ erg s$^{-1}$. The time evolution for the radio emission is also similar across the sample. The 10 GHz luminosity gradually rises, plateaus between $t_{\rm rest} \sim 50$–$100$ d after the initial discovery, and then begins to fade. During the rise, the radio emission is typically optically thick (has a positive spectral index; Figure 5). The **decay** rates of the radio

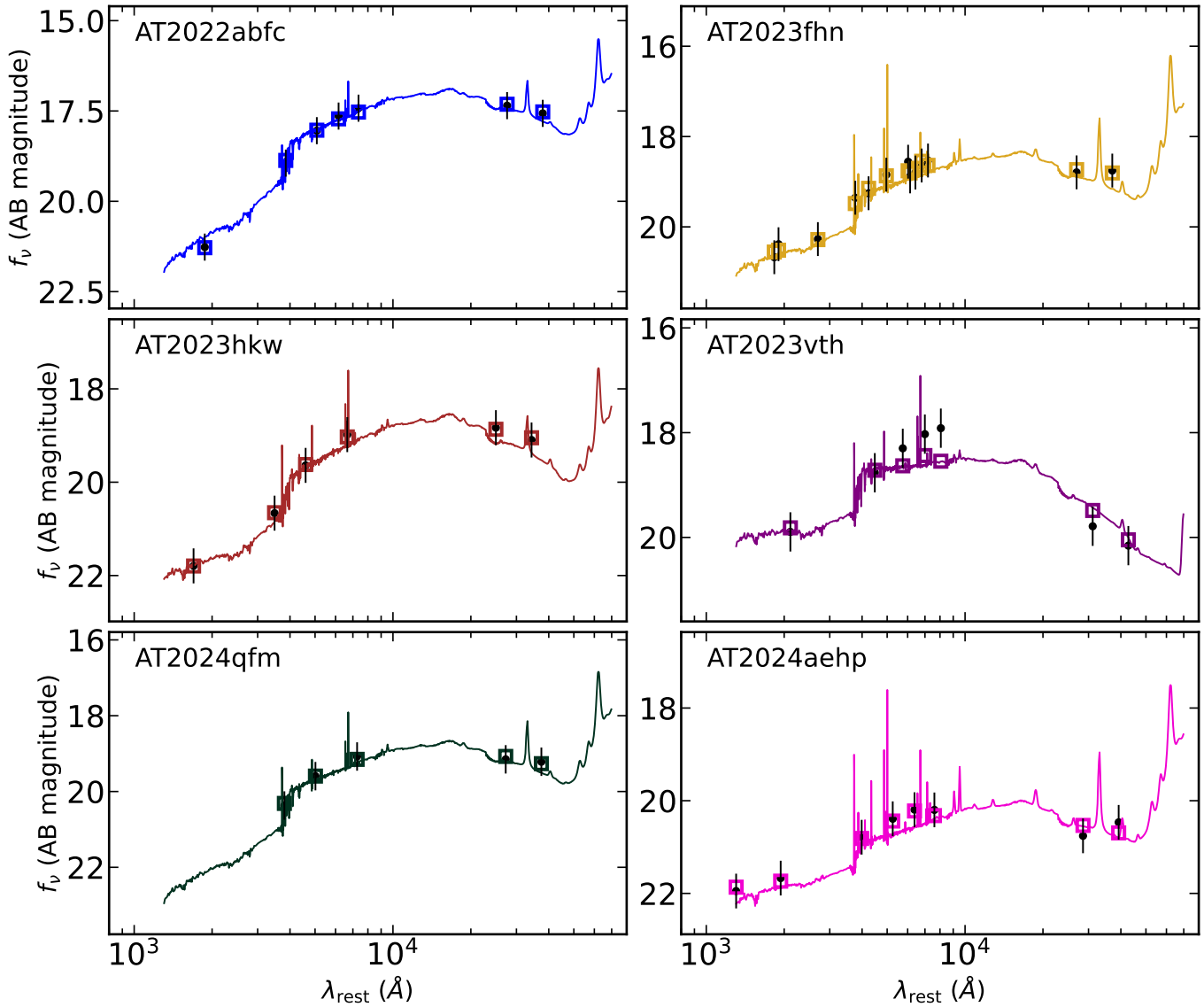


**Figure 10.** Best-fit host-galaxy spectra and photometry from `prospector` (color) as well as the input photometry (black circles with error bars). Note that the parameters reported in Table 2 are the median, 16th percentile, and 84th percentile. The best-fit host galaxy spectra displayed here corresponds to the maximum **a posteriori model**, which **may not necessarily have** the median parameters. The abscissa displays rest-frame wavelength.

emission vary, and in some cases are poorly constrained (e.g., for AT2022abfc). The fast fade of AT2023vth is similar to that of CSS161010 (Coppejans et al. 2020). **Meanwhile**, the mm-wave light curves appear to peak close to $t_{\rm rest} = 20 - 30\,$d (Figure 6). **All SEDs either have or are consistent with a broken-power law shape. The frequency is near 50 GHz during the first detections at $t_{\rm rest} \sim 30\,$d, then gradually shift to lower frequencies, passing 10 GHz at roughly $t_{\rm rest} \sim 100\,$d.**

The uniformity of the radio luminosity and evolution of all AT2018cow-like FBOTs is striking. Of course, the radio emission had to be sufficiently luminous to be detected in the first place, but we did not require the radio luminosity to brighten over time, nor was there a requirement about the timing of the peak. The radio behavior is in sharp contrast to CCSN radio light curves, which span over four orders of magnitude in luminosity, with peak times ranging from a few days to thousands of days (Bietenholz et al. 2021). **This diversity can be explained by supposing that the presence of a dense CSM that can produce bright radio emission is independent of the occurrence of a core-collapse supernova. As this is not the case for LFBOTs, we conclude that the existence of a dense CSM must be intrinsically tied to the**

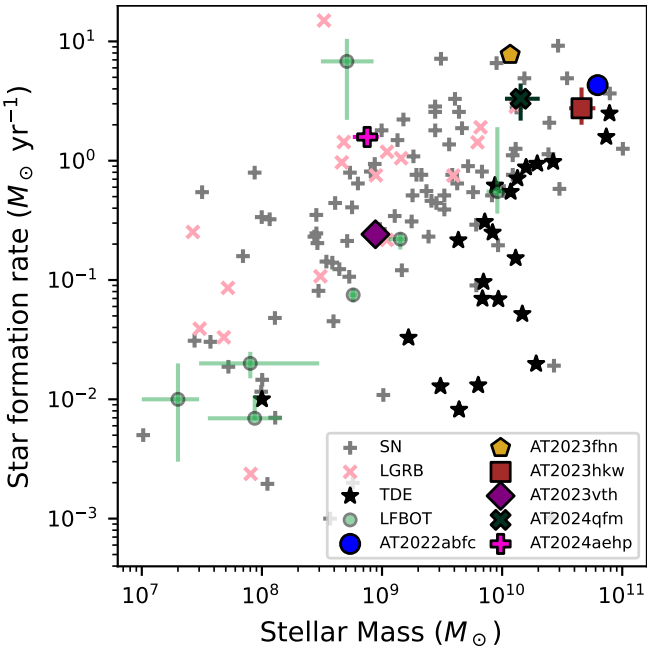


**Figure 11.** Median mass and SFR for each LFBOT's host galaxy in comparison to the host galaxies of other energetic transients. Colored lines show error bars, though they may be smaller than the marker for some points. Data for the other transients from (Law-Smith et al. 2017; French et al. 2020; Taggart & Perley 2021; Somalwar et al. 2025).

**LFBOT formation mechanism. We discuss the repurcussions of this further in Section 5.2.**

We have fewer constraints on the X-ray behavior, as several of the objects only have upper limits on their luminosity (the calculated X-ray luminosity upper limits are driven mostly by their redshift, as *Swift* determined similar photon count upper limits for each non-detection). **But as seen in Figure 3,** these upper limits appear to rule out X-ray emission similar to those of jetted TDEs or classical LGRBs while allowing for the possibility of X-ray emission similar to that of AT2018cow. **In addition**, the order-of-magnitude jump in X-ray luminosity seen in AT2024qfm over just two days is much more pronounced than the X-ray variability observed in past LFBOTs.

### 5.2. *Implications for Progenitors*

The consistent radio light curves imply consistent properties of the surrounding medium on scales of $\sim 10^{16}$–$10^{17}$ cm. **Assuming this medium was produced by some wind or mass-loss emanating from the progenitor at a speed $\lesssim 0.01c$, then it must have been emitted $\gtrsim 100$ yrs ago. Therefore, the radio behavior favors progenitor theories that can explain the presence of a dense ambient medium that is fairly similar from event to event and is produced well in advance of the explosion. This is hard to reconcile with a TDE scenario. Margutti et al. (2019) finds that the IMBH invoked to explain LFBOTs are unable to sustain outflows sufficient to produce this medium.**. Difficulties also arise in the massive-star core-collapse model: the stars that we usually suspect to be the progenitors of luminous explosions and energetic SNe are not expected to undergo significant mass loss in such a consistent way just prior to the explosion (indeed, SN radio light curves are highly diverse). On the other hand, progenitor models involving mergers (e.g., Metzger 2022; Tsuna & Lu 2025; Klencki & Metzger 2025) naturally explain the dense medium as material ejected from the merger. **Each proposed merger scenario involves mass transfer from a massive donor star to a compact object. These papers estimate the circumburst medium density from calculations of the stellar wind from the donor star and mass-loss during the merger, and have found them to be consistent with density estimates from past LFBOT papers. These estimates are also expected to hold for each instance of such a merger, explaining the consistency across the LFBOT sample.**

### 5.3. *AT2024aehp*

AT2024aehp has several characteristics that distinguish it from the other LFBOTs in this paper. Its optical light curve exhibited a plateau **at $M = -19$ mag from 5–50 d, behavior we can rule out in AT2023vth and AT2024qfm but not in AT2022abfc, AT2023fhn, and AT2023hkw. AT2024aehp is also an outlier in terms of radio evolution. As shown in Figure 4 and Figure 5, AT2024aehp dramatically brightens in radio luminosity at a later time than the other transients we have discussed, by over an order of magnitude in all frequencies between the $t_{\rm rest} \sim 70$ d and $t_{\rm rest} \sim 120$ d epochs. In contrast, the radio emission of other LFBOTs has already peaked by $t_{\rm rest} \sim 120$ d and begun to shift to lower frequencies.** Out of all the LFBOTs discussed, AT2024aehp is **the second closest** to the nucleus of its host galaxy, with a projected distance within 1 kpc. The properties of AT2024aehp are reminiscent of TDEs. TDEs **are expected to occur in the nucleus as they involve the galaxy's central supermassive black hole (Gezari 2021)**, exhibit late-time radio brightening (Horesh et al. 2021; Cendes et al. 2024), and have a slower optical light-curve decay following $t^{-5/3}$ (Evans & Kochanek 1989). While known TDEs have optical rises lasting weeks (Blagorodnova

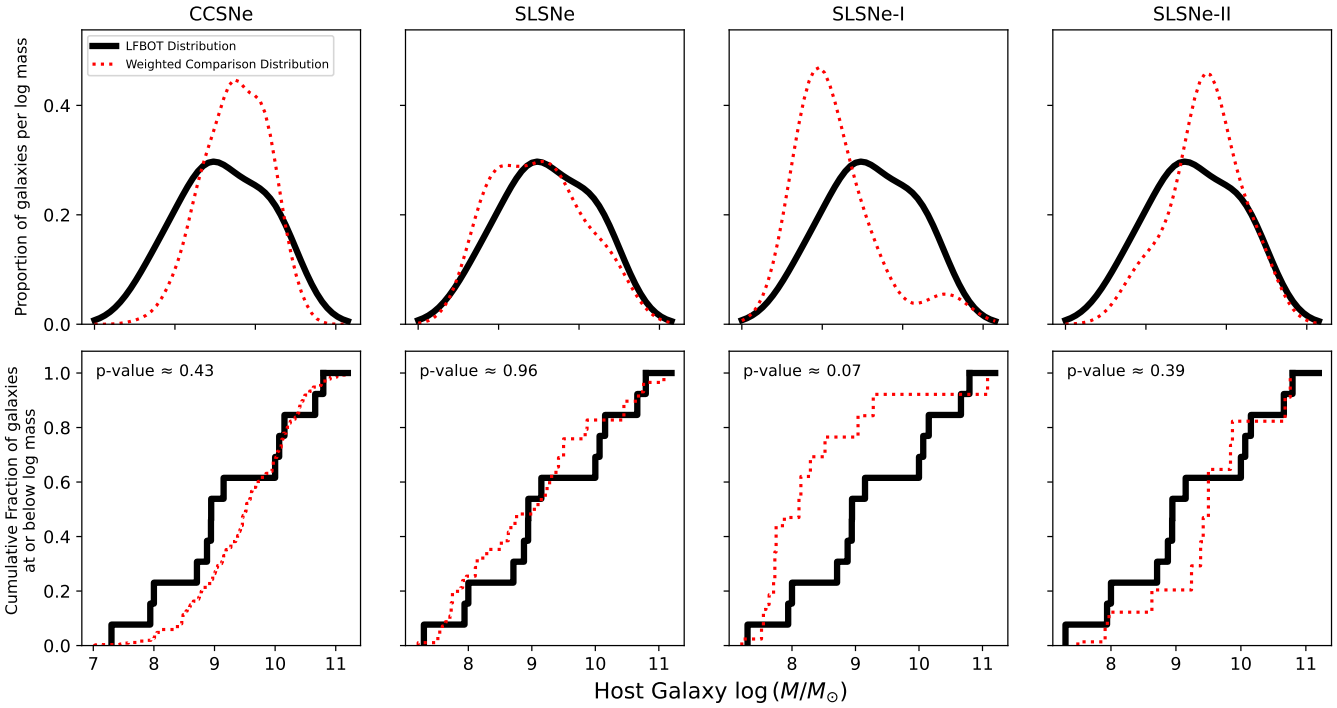


**Figure 12.** Gaussian KDEs for the LFBOT host-galaxy masses (black) in comparison to the Gaussian KDEs for each of the weighted mass distributions from the Taggart & Perley (2021) CCSN/SLSN host-galaxy population (dashed red). We also list the p-values from our Kolgomorov-Smirnov test.

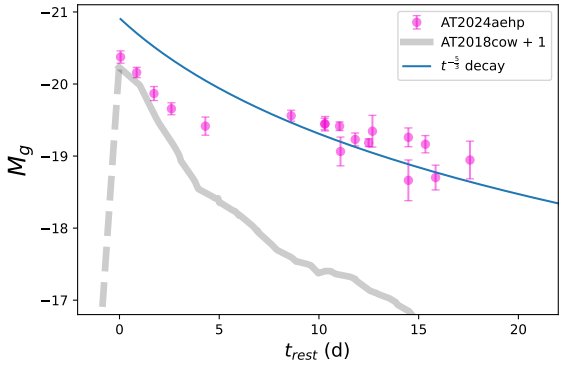


**Figure 13.** Optical $g$-band light curves for AT2024aehp (pink points) and AT2018cow (gray line). AT2018cow represents a typical LFBOT light curve. We also show a best fit of a $t^{-5/3}$ decay to the AT2024aehp light curve, as that is the canonical optical decay timescale for a TDE (Evans & Kochanek 1989). The time axis is measured in the rest frame of each transient, where $t_{\rm rest} = 0$ refers to the peak of the optical light curve.

et al. 2017), a TDE from an intermediate-mass black hole could have a much shorter rise (Kuin et al. 2019; Perley et al. 2019; Gutiérrez et al. 2024). Indeed, the transient AT2024puz—which produced luminous optical/UV evolving on a $\approx 20$ d timescale as well as luminous ($10^{44}$ erg s$^{-1}$) X-ray emission—was posited to be an object intermediate between LFBOTs and TDEs (Somalwar et al. 2025). Observations of AT2024aehp are ongoing, and a detailed analysis of this transient will be presented in future work.

### 5.4. *Considerations for Future Work*

From our work discovering and performing multiwavelength observations of LFBOTs, we have identified areas to improve on for future analysis, summarized below.

*Observations* – **While AT2024aehp appears to be an outlier, it will still be beneficial to ensure that some of its unusual characteristics are indeed unique among this class.** The optical light curve plateau of AT2024aehp motivates long-term ($> 10$ d) optical monitoring of LFBOTs. As shown in Figure 1, this extended follow-up was not present in our sample's early LFBOTs **AT2022abfc, AT2023fhn, and AT2023hkw**, which limits our ability to constrain the presence of a similar plateau. We should also **consider ensuring that our radio follow-up extends past $t_{\rm rest} \sim 120$ d rule out a late and sustained brightening such as in AT2024aehp.**

Another gap in our follow-up observations is the lack of X-ray detections for most of these LFBOTs. *Swift* observations were not sensitive enough to rule out levels of X-ray emission that have already been detected in previous LFBOTs such as CSS161010 (Coppejans

et al. 2020) and AT2018cow (Perley et al. 2019). Going forward, deeper *Chandra* observations should be obtained for each object, as was done in, e.g., AT2023fhn (Chrimes et al. 2024a). **X-ray detections will let us verify that the X-ray emission is in excess of the extrapolated synchtrotron SED, confirming that it originates from a different source, like in previous LFBOTs (e.g. Margutti et al. 2019; Bright et al. 2022). This also allows us to track the ratio of X-ray to UV/optical flux over time. Under the proposed scenario that the UV/optical flux is reprocessed high-energy radiation (Margutti et al. 2019; LeBaron et al. 2025), this would allows us to estimate the optical depth evolution of the ejecta's reprocessing layer. Finally, as the X-ray flux is thought to originate from a central object source (Ho et al. 2019; Metzger 2022; Nayana et al. 2025), it is one of the few probes of the regions closest to the transient's center.**

*Modeling* – The models we use in this paper's analysis could be improved upon or have assumptions that could be relaxed to explore a wider range of parameter space.

We use the model of Chevalier (1998) to derive properties of the shock and ambient medium from our radio and millimeter observations. This formulation requires assuming that certain conditions hold, such as the equipartition of energy, a non-thermal electron energy distribution, and that the shock is spherical and nonrelativistic. The nonrelativistic assumption will introduce some inaccuracies, as we calculate some values of $\Gamma\beta$ that are greater than 0.5. We also assume values for the filling factor $f$, $\epsilon_e$, $\epsilon_B$, and the electron power-law index, and our results do depend on these parameters. A future analysis could relax some of these assumptions or fit for parameters that we set to be fixed. Some of these flaws can be inferred from our fitting results. According to Chevalier (1998), the broken power-law fit should have a $\nu^{5/2}$ dependence in the optically thick regime and a $\nu^{-(p-1)/2}$ dependence in the optically thin regime, where $p$ is the electron power-law index. We assume $p = 3$, so the optically thin dependence should be as $\nu^{-1}$. However, for most epochs we do not have enough detections to confirm a $\nu^{5/2}$ dependence in the optically thick regime (Figure 5). In addition, from the results shown in Table 7, the observed slopes in the optically thin regime are not all consistent with $\nu^{-1}$. Work has already been done to extend this model to account for thermal electrons (Margalit & Quataert 2021) and relativistic effects (Margalit & Quataert 2024; Ferguson & Margalit 2025). Generalizing and refining the synchrotron model would help obtain more rigorous estimates of shock and ambient medium parameters. **For example, relativistic effects become significant at $\Gamma\beta \gtrsim 1$—which some of our LFBOTs approach—where results can differ by an order of magnitude (Ferguson & Margalit 2025). Our conclusions rely on the similarity of the radio emission between LFBOTs and would still be applicable, but our numerical results can be significantly affected. As our paper's scope is to present new observational data for six LFBOTs and use techniques analogous to those in previous LFBOTs papers to compare properties across this sample, we believe that work with improved synchrotron radiation models is best reserved for a future work.**

In our `prospector` fits for the LFBOT host-galaxy properties, we used a single prescription for the SFH, dust model, and initial mass function, but a future analysis should ensure that results are robust to different choices for these parameters. In particular, we use a parametric SFH to **match the work of previous LFBOT papers, but internal testing with a non-parametric SFH produced results that differ by about 1-$\sigma$.** We also omit the spectra when performing our fits, as it is difficult to standardize spectra from different observing conditions, processed through different software, and possessing different levels of contamination from the transient itself. This required us to use the mass-metallicity relation in Gallazzi et al. (2005) to fix a metallicity in our models. **As the mass is largely set by the galaxy's overall luminosity, we expect that this would lead to little change in our mass estimates, while biasing our metallicity estimates to more moderate values. We have also observed that the star-formation rate estimate is insensitive to changes in metallicity. So, our method does introduce a caveat to our metallicity estimates, but we still have confidence in our host-galaxy analysis as it relies on the mass and star-formation rate. Still,** incorporating a host-galaxy spectrum obtained after the transient faded would produce a more accurate result.

*Selection Bias* – **We make no claims that our LFBOT sample is complete or unbiased. Our requirements for LFBOT classification introduce biases that are difficult to acoount for. For example, the requirement of a visible host galaxy biases our sample towards more luminous hosts. While our analysis in Section 4.2 demonstrates some techniques we can use with a larger sample size, a proper statistical analysis and comparisons of transient populations require us to fully account for biases in the construction of our LFBOT sample.** We must also do the same for **samples**

**of other transient classes that we use as a comparison**. Ensuring that a sample is representative is especially difficult for rarer classes of transients. While we did account for some bias in our LFBOT sample in our study on host-galaxy masses, we did not perform any such correction for our analysis of host-galaxy offsets**, so our results should be taken with a grain of salt**. **Another source of bias is our requirement of an X-ray and radio detection. While this would exclude LFBOTs that are X-ray and radio faint, our procedure's success rate of detecting X-rays or radio emission from promising LFBOT candidates implies that the loss is not significant.** Representative samples of a variety of transient classes in larger numbers and a broader span of redshifts will be necessary to rigorously perform this type of analysis.

*Future Surveys* – The development of the Vera C. Rubin Observatory (Ivezić et al. 2019) and its deep, wide-field optical imaging survey is expected to increase the transient discovery rate by orders of magnitude. This would cause the number of known LFBOTs to increase significantly, paving the way for more robust and rigorous analysis. The results presented in this paper inform how to best handle each new LFBOT discovery and serve as a blueprint for statistical analysis of an expanded LFBOT sample. Furthermore, Rubin data, along with the advent of high-cadence wide-field UV surveys such as the Ultraviolet Transient Astronomy Satellite (ULTRASAT; Shvartzvald et al. 2024) and the Ultraviolet Explorer (UVEX; Kulkarni et al. 2021), will enable LFBOTs to be identified and followed up much earlier in their evolution. **Rubin's sensitivity can allow for detections before the optical peak, allowing us to identify rapid rising light curves within a couple of days. Meanwhile, ULTRASAT's rapid cadence (5 minutes) of a single field means that if we are fortuitous enough to find an LFBOT in that patch of sky, its color and fast brightening can be identified in a matter of hours.**

## 6. SUMMARY

We presented multiwavelength data—including optical photometry, optical spectroscopy, *Swift* X-ray observations, and VLA/NOEMA/uGMRT radio observations—of six LFBOTs (AT2022abfc, AT2023fhn, AT2023hkw, AT2023vth, AT2024qfm, and AT2024aehp) at redshifts spanning $z = 0.0747$–$0.339$. We identified all of these objects on the basis of fast light-curve evolution in ZTF survey data, and confirmed their similarity to the prototype AT2018cow via radio detections. Five of the six LFBOTs (all except AT2023fhn) are presented here for the first time. Our work increases the known number of LFBOTs by 50%.

The optical light curves of four events fade by 3–4 mag over $t_{\rm rest} \approx 10$ d, similarly to AT2018cow. A fifth event, AT2024aehp, exhibited a plateau at $M \approx -19$ mag over tens of days, after an initial fast decline. The optical spectra do not show any clear features from the transients themselves, including a spectrum of AT2024aehp during its plateau phase.

The X-ray luminosity of AT2024qfm is among the highest of all LFBOTs to date, at $10^{44}\,{\rm erg\,s^{-1}}$ at $t_{\rm rest} \approx 1$ week. The X-rays also appeared to show a brightening by an order of magnitude over just a few days. *Swift* X-ray observations of the other objects resulted in non-detections that were not deep enough to rule out emission identical to AT2018cow itself. AT2023fhn was detected by Chandra at similar levels to AT2018cow (Chrimes et al. 2024b; Nayana et al. 2025).

The 10 GHz radio light curves peak at $t_{\rm rest} \approx 50$–$100$ d, with peak luminosities primarily in the range of $10^{39}$–$10^{40}\,{\rm erg\,s^{-1}}$. During the rise, the emission appears to be self-absorbed, confirmed by multi-band VLA observations for all objects, as well as by NOEMA 100 GHz detections for AT2024qfm and AT2023vth. We measured basic shock parameters using a simple equipartition analysis, and found—as for previous LFBOTs—fast but sub-relativistic shock speeds. The exception to the radio behavior is AT2024aehp, which, starting at $t_{\rm rest} \approx 70$ d, displayed rapid brightening. This late-time brightening is reminiscent of TDEs. Observations of AT2024aehp are ongoing, and will be presented in future work.

For all objects, we measured the offset of the transient position from the host-galaxy nucleus, as well as the basic host-galaxy properties (e.g., stellar mass, star formation rate or SFR). Combining our results with all literature objects, for a total sample of 13 objects, we find that the offset distribution appears intermediate to CCSNe and SLSNe. We find that the host galaxy masses and SFRs trace the full CCSN sequence, and that—even correcting for the fact that we are biased towards detecting LFBOTs in higher-mass galaxies, due to our requirement of a host-galaxy detection—the intrinsic mass distribution appears to be consistent with both SLSNe and CCSNe.

The similar radio light curves imply a similar circumburst medium for LFBOTs, and therefore a progenitor scenario that can produce a consistent medium timed with the terminal event, such as the merger of a massive star with a compact object. This argument was made when the sample size of LFBOTs was much smaller (Metzger 2022), but here we confirm it with

a much larger number of events. **For future LFBOT studies, we would suggest** longer-term optical monitoring (to confirm or rule out the presence of a plateau as observed in AT2024aehp), deeper X-ray observations (i.e., with *Chandra*), and selection in other bands of the electromagnetic spectrum to mitigate some of the biases present here. **LFBOT studies are also poised to benefit from new and upcoming surveys. The deep sensitivity of the Legacy Survey of Space and Time on the Rubin Observatory and the devlopment of very-high-cadence surveys like ULTRASAT can simultaneously expand our number of LFBOTs and enable us to discover them earlier in their evolution. Faster discoveries enable earlier radio observations at $t_{\rm obs} \leq 5$ days, probing smaller shock radii and more recent mass-loss history.**

## 7. DATA AVAILABILITY

**All data and code underlying this paper are available at https://github.com/CassieSev/Multiwavelength-Analysis-of-Six-LFBOTs.**

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authors wish to recognize and acknowledge the very significant cultural role and reverence that the summit of Maunakea has always had within the indigenous Hawaiian community. We are most fortunate to have the opportunity to conduct observations from this mountain.

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These results made use of the Lowell Discovery Telescope (LDT) at Lowell Observatory. Lowell is a private, nonprofit institution dedicated to astrophysical research and public appreciation of astronomy and operates the LDT in partnership with Boston University, the University of Maryland, the University of Toledo, Northern Arizona University and Yale University. The Large Monolithic Imager was built by Lowell Observatory using funds provided by the U.S. NSF (AST-1005313). This paper includes data gathered with the 6.5 m Magellan Telescopes located at Las Campanas Observatory, Chile. Based in part on observations obtained at the Southern Astrophysical Research (SOAR) telescope, which is a joint project of the Ministério da Ciência, Tecnologia e Inovações (MCTI/LNA) do Brasil, the U.S. NSF's NOIRLab, the University of North Carolina at Chapel Hill (UNC), and Michigan State University (MSU).

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Legacy Survey (MzLS; Prop. ID #2016A-0453; PI A. Dey). DECaLS, BASS and MzLS together include data obtained, respectively, at the Blanco telescope, Cerro Tololo Inter-American Observatory, NSF's NOIRLab; the Bok telescope, Steward Observatory, University of Arizona; and the Mayall telescope, Kitt Peak National Observatory, NOIRLab. Pipeline processing and analyses of the data were supported by NOIRLab and the Lawrence Berkeley National Laboratory (LBNL). The Legacy Surveys project is honored to be permitted to conduct astronomical research on Iolkam Du'ag (Kitt Peak), a mountain with particular significance to the Tohono O'odham Nation.

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# APPENDIX

## A. IDENTIFICATION **AND CLASSIFICATION**

**In this section, we describe how each LFBOT was identified in the ZTF alert stream, and describe the initial follow-up observations that led to their classification as an LFBOT.** ZTF observations of the six LFBOTs in this paper occurred during Phase II (ZTF-II), which was divided into public (50%), partnership (30%), and Caltech (20%) observing time. ZTF uses three filters: *ztf-g, ztf-r*, and *ztf-i*. For comparison with other photometry, we denote these as *g, r*, and *i*. ZTF data products are processed by the IPAC ZTF pipeline (Masci et al. 2019), which uses image subtraction following the methods of Zackay et al. (2016). Detections more significant that 5-$\sigma$ are distributed as alerts in the Avro format (Patterson et al. 2019). These alerts are filtered using machine-learning real-bogus metrics (Mahabal et al. 2019; Duev et al. 2019), host-galaxy classifiers (Tachibana & Miller 2018), and inspection of light-curve properties. For ZTF-II, the collaboration used the Fritz marshal (van der Walt et al. 2019; Duev et al. 2019) to coordinate detection, analysis, and follow-up observations of transients. The Fritz marshal uses the open-source software package `SkyPortal` (Coughlin et al. 2023).

### A.1. *AT2022abfc/ZTF22abvrxkk*

AT2022abfc was first detected by ZTF on November 21, 2022 (MJD = 59904.344; Fremling 2022). ZTF non-detections prior to the first observation and ZTF detections over the next five days confirmed a rise faster than 1 mag day$^{-1}$ and a fade rate of 0.2 mag day$^{-1}$. The transient's fast evolution led to follow-up multiwavelength observations. A Gemini South spectrum taken December 1, 2022 showed narrow H$\alpha$ emission and Ca II H&K absorption lines at $z = 0.212$, implying a peak absolute magnitude of $-20.7$ (Ho et al. 2022a). A radio follow-up campaign with the VLA began with a detection at 10 GHz on December 6 (Ho et al. 2022a). Subsequent radio observations showed a brightening of the radio emission. The transient's fast evolution, high luminosity, and radio emission led to its classification as an LFBOT.

### A.2. *AT2023fhn/ZTF23aaeozpp*

The discovery narrative for AT2023fhn has already been published by Chrimes et al. (2024b); we reiterate it here for convenience. ZTF first detected AT2023fhn on April 10, 2023 (MJD = 60044.204; Fremling 2023a). Its light curve showed a fast fade of 0.2 mag day$^{-1}$ and blue optical colors ($g - r \approx -0.5$ mag). A possible host galaxy's photometric redshift implied an absolute optical magnitude of $M_g < -20$, motivating follow-up observations. On April 20, a spectrum obtained by the DBSP on the Palomar 5 m telescope yielded H$\alpha$, N II, and S II emission lines at $z = 0.24$, confirming an absolute magnitude of $M_g = -21.5$ (Ho et al. 2023a). The transient was subsequently detected at X-ray and radio wavelengths, respectively by *Chandra* on April 25 (Chrimes et al. 2023) and the VLA (10 GHz) on June 15 (Ho 2023a). The blue colors, fast fading, high luminosity, and detected emission throughout the electromagnetic spectrum confirmed it as an LFBOT. The *Chandra* X-ray detections and a subset of this transient's VLA radio observations are already published by Chrimes et al. (2024a).

### A.3. *AT2023hkw/ZTF23aaimsja*

ZTF first detected AT2023hkw on May 1, 2023 (MJD = 60065.199; Fremling 2023b). The transient was flagged for fast evolution from a rise rate greater than 1 mag day$^{-1}$ and a fade rate of $\sim$ 0.2 mag day$^{-1}$. This resulted in follow-up spectroscopy on May 12 using DEIMOS at the Keck Observatory (Li et al. 2023), using 1440 s of exposure time. The spectrum showed narrow H$\alpha$ and [N II] emission lines from the host galaxy at $z = 0.339$, implying an absolute magnitude of $M_g = -21$. It was also detected by the VLA at 10 GHz on June 14 (Ho 2023b). It was classified as an LFBOT from its luminosity, fast evolution, and radio emission.

### A.4. *AT2023vth/ZTF23ableqsp*

ZTF first detected AT2023vth on October 18, 2023 (MJD = 60235.116; Fremling 2023c). It had a fast rise of 1 mag day$^{-1}$, a fast fade of 0.25 mag day$^{-1}$, and blue colors with $g - r \approx -0.4$ mag (Sevilla et al. 2023), making it a promising LFBOT candidate. Optical spectroscopy taken on November 12 revealed narrow H$\alpha$, [N II], and [S II] emission lines from the host galaxy at $z = 0.0747$, which implied an absolute magnitude of $M_g \approx -20$ (Ho et al. 2023a). The VLA detected radio emission at 10 GHz on November 17 (Ho 2023c) and NOEMA found millimeter emission at 100 GHz on November 20 (Ho & Bremer 2023). The presence of multiwavelength emission solidified the transient's LFBOT classification. We also obtained deep late-time VLT observations that will be reported in a separate paper **according to a prior arrangement** (Wise et al., in prep.).

### A.5. *AT2024qfm/ZTF24aaxhxhf*

AT2024qfm was first detected by ZTF on July 24, 2024 (MJD = 60515.41; Forster et al. 2024). The most recent non-detection was 0.98 d prior at $g > 19.27$ mag. The source was detected again three nights later as part of the ZTF public survey, resulting in the first alert. Over the following two nights the source exhibited significant fading of 0.3 mag day$^{-1}$. SDSS pre-imaging showed a galaxy at the transient's position with photometric redshift $z = 0.190 \pm 0.054$, implying a high luminosity of $M < -20$ mag. The rapid fading, host-galaxy counterpart, and high luminosity led the transient to be flagged by the daily scanner on July 27.

The rapid fading flagged several fast-transient pipelines including Fastfinder (Fulton et al. 2024) and ZTFReST (Andreoni et al. 2021). The rapid fading was publicly reported by Fastfinder (Fulton et al. 2024). The transient itself was reported to the Transient Name Server (TNS[45]) by the Automatic Learning for the Rapid Classification of Events (ALeRCE) broker (Forster et al. 2024).

On August 1, a redshift of $z = 0.2270$ was determined by Gillanders et al. (2024) and an X-ray detection was reported by Margutti et al. (2024). On August 2, we obtained a long-slit spectrum of AT 2024qfm with GMOS on Gemini-North under a ToO program[46] and confirmed the reported redshift. A VLA radio observation was also obtained **(Ho 2024)** ; the radio data will be published as part of a separate paper (Nayana A. J. et al., in prep.). Finally, we obtained 100 GHz observations using NOEMA, which resulted in a detection. The fast optical light curve, and high luminosities at millimeter and X-ray wavelengths, all confirmed the LFBOT nature of this source. We also obtained deep late-time GTC/HiPERCAM observations that will be reported in a separate paper (Wise et al., in prep.).

### A.6. *AT2024aehp/ZTF24abygbss*

AT2024aehp was first detected by ZTF on December 19, 2024 (MJD = 60663.36; Munoz-Arancibia et al. 2024). The most recent non-detection was 3 d prior. ZTF photometry over the next two nights revealed a fade rate of 0.2 mag day$^{-1}$, causing the transient to be flagged by the daily scanner on December 21. The fast fading, **blue colors ($g - r \approx -0.3$ mag)** and possible high peak luminosity from the photometric redshift of the host galaxy (SDSS $z_{\rm ph} = 0.17 \pm 0.09$) motivated us to trigger follow-up observations. We acquired spectroscopy of the candidate using GMOS on Gemini South on December 23, 2024 The spectrum shows narrow H$\alpha$, [N II], and [S II] emission lines that correspond to $z = 0.170$. We reported the rapid evolution and the high luminosity **inferred from the spectroscopy.** (Sevilla et al. 2024).

Radio and X-ray observations initially resulted in non-detections, and the optical light curve flattened out in a way uncharacteristic of LFBOTs (Schroeder et al. 2025). A subsequent radio epoch at 86 days after peak light resulted in a detection and confirmation of a positive spectral index, suggesting a link with AT2018cow-like LFBOTs (Schroeder & Ho 2025). Then, another epoch at 141 days showed brightening in all radio bands by almost an order of magnitude, which is unprecedented compared to the radio emission of other LFBOTs. This paper focuses on the early observations of AT2024aehp, and the interesting late-time behavior will be examined in separate work.

## B. DATA TABLES

In this section we provide tables of optical spectroscopy (Table 3), X-ray observations (Table 4), radio observations (Table 5), and host-galaxy photometry (Table 6). We also give our broken power-law fits to the radio SEDs (Table 7) and the resulting synchrotron parameters (Table 8). Finally, we list the optical photometry for each transient in Table 9.

[45] https://www.wis-tns.org/
[46] GN-2024B-Q-128; PI A. Ho

**Table 3.** Optical Spectroscopy

| Object | UTC Date | $t_{\rm obs}$ (days) | Telescope/Inst. | Exposures | PI & Program | Reduction Software |
|---|---|---|---|---|---|---|
| AT2022abfc | 20221201 | 10 | Gemini-S/GMOS | $6 \times 450$ s | GS-2022B-Q-126; PI A. Ho | DRAGONS* |
| AT2023fhn | 20230419 | 7 | Gemini-S/GMOS | $6 \times 450$ s | GS-2023A-Q-127; PI A. Ho | DRAGONS* |
| AT2023fhn | 20230420 | 8 | P200/DBSP | $3 \times 600$ s | PI Y. Qin | DBSP_DRP** |
| AT2023fhn | 20230426 | 14 | Keck-I/LRIS | $4 \times 2700$ s | PI M. M. Kasliwal | lpipe*** |
| AT2023hkw | 20230512 | 12 | Keck-II/DEIMOS | 1440s | PI A. Filippenko | IRAF routines and custom codes † |
| AT2023vth | 20231022 | 2 | P200/SEDM | $\sim 1800$ s†† | —†† | —†† |
| AT2023vth | 20231112 | 23 | Gemini-N/GMOS | $2 \times 500$ s | GN-2023B-Q-130; PI A. Chrimes | DRAGONS* |
| AT2024qfm | 20240802 | 6 | Gemini-N/GMOS | $4 \times 450$ s | GN-2024B-Q-128; PI A. Ho | DRAGONS* |
| AT2024qfm | 20240809 | 13 | Keck-I/LRIS | $4 \times 875$ s | U256; PI Chornock | lpipe*** |
| AT2024qfm | 20240831 | 35 | MMT/Binospec | 1200 s | UAO-G200-24B; PI A. A. Miller | Binospec and pypeit††† |
| AT2024qfm | 20240902 | 37 | Keck-I/LRIS | $4 \times 900$ s | U233; PI Filippenko | lpipe*** |
| AT2024aehp | 20241224 | 5 | Gemini-S/GMOS | $4 \times 450$ s | GS-2024B-Q-121 ; PI A. Ho | DRAGONS* |
| AT2024aehp | 20250101 | 13 | Keck-II/KCWI | $600 + 3 \times 180$ s | PI Y. Qin | kskywizard†††† |
| AT2024aehp | 20250107 | 19 | MMT/Binospec | $4 \times 500$ s | PI A. Gal-Yam | Binospec and pypeit††† |

Note—*: DRAGONS (Labrie et al. 2023) is a software package specialized for reducing Gemini data, and performs stacking of flatfields and comparison lamps, bias subtraction, and standard-star flux calibration.
**: DBSP_DRP (Mandigo-Stoba et al. 2022) is a Python package based on pypeit (Prochaska et al. 2020) that performs automated reduction of DBSP data, including atmospheric corrections andflux calibration.
***lpipe (Perley 2019) is an IDL pipeline that performs for fully automated data reduction for the LRIS instrument. Its steps include bias correction, flat-fielding, and wavelength and flux calibration.
†: The IRAF routines (Silverman et al. 2012; Tody 1986)[a] routines and custom Python and IDL codes [b] use techniques for CCD processing and spectrum extraction, comparison-lamp wavelength calibration, and standard star flux calibration.
††: This spectrum was taken as part of automated follow-up for the ZTF Bright Transient Survey (Fremling et al. 2020).
†††: A Binospec pipeline (Kansky et al. 2019) was used to perform bias subtraction and flat fielding, while pypeit (Prochaska et al. 2020) was used to perform cosmic-ray removal, wavelength calibration and relative-flux calibration from a standard star.
††††: kskywizard[c] is a pipeline that can perform sky subtraction, flux calibration, and telluric correction.

[a] https://iraf-community.github.io
[b] https://github.com/ishivvers/TheKastShiv
[c] https://github.com/zhuyunz/KSkyWizard

**Table 4.** *Swift* XRT Observations

| Object | UTC Date | $t_{\rm obs}$ (days) | Flux ($10^{-13}$ erg cm$^{-2}$ s$^{-1}$) | Count Rate ($10^{-3}$ s$^{-1}$) | $n_{\rm H}$ ($10^{20}$ cm$^{-2}$) | $L_{\rm X}$ ($10^{43}$ erg s$^{-1}$) | Photon Index |
|---|---|---|---|---|---|---|---|
| AT2022abfc | 20221208 | 17 | $< 2$ | $< 6$ | 2.49 | $< 3$ | — |
| AT2023fhn | 20230425 | 13 | $< 1.7$ | $< 5$ | 2.26 | $< 4$ | — |
| AT2023hkw | 20230521 | 21 | $< 1.1$ | $< 3.5$ | 1.05 | $< 4.5$ | — |
| AT2023vth | 20231110 | 21 | $< 1.9$ | $< 3.4$ | 22.6 | $< 0.23$ | — |
| AT2024qfm | 20240801 | 5 | $5.0 \pm 0.8$ | $13 \pm 2$ | 3.83 | $8.2 \pm 1.3$ | $2.1^{+0.8}_{-0.5}$ |
| AT2024qfm | 20240802 | 6 | $3.7 \pm 0.9$ | $10 \pm 2$ | — | $6.1 \pm 1.5$ | $1.5^{+1.6}_{-0.9}$ |
| AT2024qfm | 20240803 | 7 | $< 4.9$ | $< 12$ | — | $< 7.9$ | — |
| AT2024qfm | 20240803 | 7.5 | $2.8^{+1.8}_{-1.3}$ | $7^{+5}_{-3}$ | — | $4.5^{+2.9}_{-2.1}$ | $2.9^{+3.3}_{-2.9}$ |
| AT2024qfm | 20240805 | 9 | $< 3.1$ | $< 8$ | — | $< 5.0$ | — |
| AT2024qfm | 20240807 | 11 | $25^{+24}_{-14}$ | $63^{+60}_{-35}$ | — | $40^{+38}_{-23}$ | — * |
| AT2024qfm | 20240809 | 13 | $< 23$ | $< 58$ | — | $< 37$ | — |
| AT2024qfm | 20240809 | 13.5 | $2.0^{+1.2}_{-0.9}$ | $5^{+3.2}_{-2.4}$ | — | $3.3^{+2.0}_{-1.4}$ | — * |
| AT2024qfm | 20240810 | 14 | $1.7^{+1.0}_{-0.7}$ | $4^{+1.9}_{-2.7}$ | — | $2.7^{+1.6}_{-1.2}$ | $-0.3^{+1.5}_{-1.8}$ |
| AT2024qfm | 20240813 | 17 | $2.6^{+1.5}_{-1.1}$ | $7^{+4}_{-3}$ | — | $4.2^{+2.5}_{-1.9}$ | — * |
| AT2024qfm | 20241204 | 130 | $< 0.9$ | $< 2$ | — | $< 1.5$ | — |
| AT2024aehp | 20241231 | 12 | $< 2.3$ | $< 6.2$ | 2.83 | $< 2.0$ | — |
| AT2024aehp | 20250103 | 15 | $< 3.0$ | $< 7.9$ | — | $< 2.6$ | — |
| AT2024aehp | 20250105 | 17 | $< 2.4$ | $< 6.3$ | — | $< 2.1$ | — |
| CSS161010 | 20161103 | 24 | $1.8^{+1.4}_{-0.9}$ | $4.5^{+3.4}_{-2.3}$ | 4.7 | $4.8^{+3.6}_{-2.4} \times 10^{-2}$ | — * |
| CSS161010 | 20161108 | 29 | $< 1.2$ | $< 2.9$ | — | $< 3.4 \times 10^{-2}$ | — |
| CSS161010 | 20161111 | 32 | $< 2.0$ | $< 5.0$ | — | $< 5.7 \times 10^{-2}$ | — |

Note—AT2022abfc values are from Ho et al. (2022a), AT2023fhn measurements obtained from Chrimes et al. (2024b), $n_{\rm H}$ for CSS161010 obtained from Coppejans et al. (2020). To convert flux to count rate, we assume a photon power-law index of $\Gamma = 2$ for all measurements. $t$ is determined from the middle of the observation period. $n_{\rm H}$ is the same for all observations of a given object.
*Too few counts for a spectral index.

**Table 5**. Radio and Millimeter Observations

| Name | UTC Date | $t_{\rm obs}$ (days) | Band | $\nu$ (GHz) | Config. | $f_\nu$ ($\mu$Jy) |
|---|---|---|---|---|---|---|
| AT2022abfc | 20221206 | 15 | X | 10 | VLA-C | $< 13$ |
| AT2022abfc | 20221206 | 15 | X | 9 | VLA-C | $< 17$ |
| AT2022abfc | 20221206 | 15 | X | 11 | VLA-C | $< 19$ |
| AT2022abfc | 20221223 | 32 | Ku | 10 | VLA-C | $52 \pm 3$ |
| AT2022abfc | 20221223 | 32 | X | 10 | VLA-C | $38 \pm 4$ |
| AT2022abfc | 20230110 | 53 | 1 | 86 | NOEMA-12A | $< 348$ |
| AT2022abfc | 20230405 | 135 | Ku | 10 | VLA-B | $45 \pm 7$ |
| AT2022abfc | 20230405 | 135 | X | 10 | VLA-B | $43 \pm 9$ |
| AT2022abfc | 20230602 | 193 | 3 | 0.45 | GMRT | $< 150$ |
| AT2022abfc | 20230622 | 214 | 5 | 1.37 | GMRT | 44±12* |
| AT2022abfc | 20230623 | 215 | 3 | 0.45 | GMRT | $< 165$ |
| AT2022abfc | 20230624 | 216 | Ku | 10 | VLA-BnA− >A | $< 27$ |
| AT2022abfc | 20230624 | 216 | S | 3 | VLA-BnA− >A | $< 26$ |
| AT2022abfc | 20230624 | 216 | X | 10 | VLA-BnA− >A | $< 26$ |
| AT2022abfc | 20240202 | 438 | X | 10 | VLA-C | $< 12$ |
| AT2022abfc | 20240202 | 438 | X | 9 | VLA-C | $< 17$ |
| AT2022abfc | 20240202 | 438 | X | 11 | VLA-C | $< 19$ |
| AT2023fhn | 20230422 | 10 | 1 | 86 | NOEMA-C | $< 90$ |
| AT2023fhn | 20230511 | 29 | X | 9 | VLA-B | $< 19$ |
| AT2023fhn | 20230511 | 29 | X | 11 | VLA-B | $< 20$ |
| AT2023fhn | 20230615 | 64 | X | 9 | VLA-BnA | $43 \pm 6$ |
| AT2023fhn | 20230615 | 64 | X | 11 | VLA-BnA | $53 \pm 7$ |
| AT2023fhn | 20230709 | 88 | Ku | 10 | VLA-A | $75 \pm 7$ |
| AT2023fhn | 20230709 | 88 | X | 10 | VLA-A | $118 \pm 6$ |
| AT2023fhn | 20230825 | 135 | Ku | 10 | VLA-A | $98 \pm 9$ |
| AT2023fhn | 20230825 | 135 | X | 10 | VLA-A | $143 \pm 7$ |
| AT2023fhn | 20230910 | 151 | 4 | 0.65 | GMRT | $< 300$ |
| AT2023fhn | 20230910 | 151 | 5 | 1.37 | GMRT | $< 180$ |
| AT2023fhn | 20231026 | 197 | 5 | 1.37 | GMRT | $85 \pm 24$ |
| AT2023fhn | 20231103 | 205 | 4 | 0.65 | GMRT | $< 231$ |
| AT2023fhn | 20231127 | 229 | 4 | 0.65 | GMRT | $< 210$ |
| AT2023fhn | 20231127 | 229 | 5 | 1.37 | GMRT | $112 \pm 20$ |
| AT2023fhn | 20240624 | 440 | X | 10 | VLA-B | $33 \pm 5$ |
| AT2023hkw | 20230614 | 44 | X | 9 | VLA-BnA | $79 \pm 6$ |
| AT2023hkw | 20230614 | 44 | X | 11 | VLA-BnA | $103 \pm 7$ |
| AT2023hkw | 20230704 | 64 | Ka | 33 | VLA-A | $124 \pm 11$ |
| AT2023hkw | 20230704 | 64 | Ku | 10 | VLA-A | $152 \pm 7$ |
| AT2023hkw | 20230826 | 117 | C | 6 | VLA-A | $74 \pm 7$ |
| AT2023hkw | 20230826 | 117 | X | 10 | VLA-A | $85 \pm 8$ |

NOTE—**The full set of radio and millimeter observations for each LFBOT are available online in a machine-readable table. These tables include comments for specific observations** Listed frequencies are in the observer frame. We require SNR $> 3$ for a detection. Otherwise, we list the 3-$\sigma$ upper limits for a non-detection.
*This measurement was binned together with a GMRT Band-5 observation on June 2, 2023.

**Table 6.** Host-Galaxy Photometry

| Object | *GALEX* | | Legacy/Pan-STARRS | | | | *NEOWISE* | | |
|---|---|---|---|---|---|---|---|---|---|
| | FUV | NUV | $g$ | $r$ | $i$ | $z$ | $w1$ | $w2$ | $w3$ |
| AT2022abfc | — | 21.550±0.310 | 19.057±0.002 | 18.124±0.001 | 17.698±0.001 | 17.464±0.002 | 17.356±0.007 | 17.568±0.015 | 15.875±0.073 |
| AT2023fhn | — | 20.913±0.200 | 19.452±0.006 | 18.912±0.006 | 18.606±0.009 | 18.566±0.009 | 18.796±0.024 | 18.755±0.051 | — |
| AT2023hkw | — | 21.898±0.086 | 20.706±0.014 | 19.669±0.012 | — | 18.996±0.010 | 18.833±0.022 | 19.097±0.051 | — |
| AT2023vth | — | 21.601±0.274 | 19.430±0.019 | 18.774±0.012 | 18.370±0.008 | 18.179±0.014 | 19.809±0.192 | 20.170±0.380 | — |
| AT2024qfm | — | — | 20.530±0.007 | 19.706±0.006 | — | 19.135±0.007 | 19.156±0.028 | 19.218±0.067 | — |
| AT2024aehp | 22.158±0.181 | 21.902±0.115 | 20.881±0.009 | 20.454±0.009 | 20.240±0.013 | 20.232±0.020 | 20.761±0.117 | 20.468±0.206 | — |

Note—Photometry used in our `prospector` host-galaxy fits. Empty columns represent non-detections or detections with SNR $< 5$. The middle columns represent photometry in the Legacy Survey *griz* filters, or the Pan-STARRS *griz* filters **for AT2023vth as its host was outside the Legacy Survey footprint**. Not listed is *HST* photometry for AT2023fhn from Chrimes et al. (2024a) that was also used in the fit.

**Table 7.** Radio SED Broken Power-law Fit Parameters

| Object | Observer Frame Epoch (days) | Rest Frame Epoch (days) | $\nu_{\rm obs}$ (GHz) | $F_{\rm obs}$ ($\mu$Jy) | $a_2$ |
|---|---|---|---|---|---|
| AT2023fhn | 88–96 | 71–77 | $4.0 \pm 0.3$ | $232 \pm 14$ | $-0.69 \pm 0.08$ |
| AT2023fhn | 137–138 | 110–111 | $5.5 \pm 0.3$ | $387 \pm 14$ | $-0.87 \pm 0.08$ |
| AT2023vth | 86–87 | 80–81 | $8.5 \pm 0.6$ | $(2.6 \pm 0.2) \times 10^3$ | $-1.06 \pm 0.06$ |
| AT2023vth | 104–129 | 97–120 | $2.7 \pm 0.2$ | $(1.50 \pm 0.13) \times 10^3$ | $-0.55 \pm 0.04$ |
| AT2024aehp | 140–155 | 120–132 | $12.8 \pm 1.1$ | $(7.8 \pm 0.9) \times 10^2$ | $-0.67 \pm 0.09$ |

Note—Results of a broken power-law fit (Equation 7). The values for the fitted peak frequency and flux density is measured in the observer's frame.

**Table 8.** Inferred Parameters from Synchrotron Self-Absorption Modeling

| Object | $t_{\rm rest}$ (days) | $\nu_{\rm p}$ (rest) (GHz) | $R$ ($10^{16}$ cm) | $B$ (G) | $v/c$ | Energy ($10^{48}$ erg) | $n_e$ ($10^3$ cm$^{-3}$) | $\dot{M}$ ($10^{-4}$ $M_\odot$/yr) | $m_p n_e R^3$ ($10^{-5}$ $M_\odot$) |
|---|---|---|---|---|---|---|---|---|---|
| AT2022abfc | 26 | $> 19$ | $< 1.4$ | $> 1.49$ | $< 0.21$ | $< 1.7$ | $> 5.2$ | $> 0.35$ | $> 1.3$ |
| AT2023fhn | 73 | $5.0 \pm 0.4$ | $13.0 \pm 1.1$ | $0.32 \pm 0.03$ | $0.57 \pm 0.05$ | $55 \pm 6$ | $0.022 \pm 0.007$ | $0.12 \pm 0.02$ | $4.1 \pm 1.7$ |
| AT2023fhn | 111 | $6.8 \pm 0.4$ | $12.0 \pm 0.7$ | $0.41 \pm 0.02$ | $0.38 \pm 0.02$ | $74 \pm 5$ | $0.10 \pm 0.03$ | $0.47 \pm 0.06$ | $15.2 \pm 4.5$ |
| AT2023hkw | 33 | $> 15$ | $< 4.2$ | $> 0.95$ | $< 0.44$ | $< 16$ | $> 0.40$ | $> 0.22$ | $> 2.4$ |
| AT2023hkw | 87 | $> 8.0$ | $< 9.2$ | $> 0.5$ | $< 0.37$ | $< 47$ | $> 0.16$ | $> 0.43$ | $> 10.3$ |
| AT2023vth | 81 | $9.1 \pm 0.7$ | $7.1 \pm 0.6$ | $0.58 \pm 0.04$ | $0.32 \pm 0.03$ | $31 \pm 3$ | $0.31 \pm 0.09$ | $0.49 \pm 0.07$ | $9.4 \pm 3.6$ |
| AT2023vth | 110 | $2.9 \pm 0.2$ | $17.6 \pm 1.3$ | $0.19 \pm 0.01$ | $0.53 \pm 0.04$ | $50 \pm 6$ | $0.010 \pm 0.003$ | $0.10 \pm 0.01$ | $4.7 \pm 1.6$ |
| AT2024qfm | 10 | $> 115$ | $< 0.3$ | $> 8.32$ | $< 0.13$ | $< 0.58$ | $> 449$ | $> 1.48$ | $> 1.3$ |
| AT2024aehp | 73 | $> 18$ | $< 1.5$ | $> 1.36$ | $< 0.08$ | $< 1.7$ | $> 30$ | $> 2.22$ | $> 9.2$ |
| AT2024aehp | 127 | $15.0 \pm 1.3$ | $5.5 \pm 0.5$ | $0.91 \pm 0.08$ | $0.16 \pm 0.02$ | $34 \pm 5$ | $3.1 \pm 1.1$ | $3.0 \pm 0.5$ | $44 \pm 20$ |

Note—The shock parameters we derive using Equations 2–6 for the various epochs of radio observations of each LFBOT. When we can only constrain the peak of the radio SED by observing one side of the broken-power law, we calculate limits on these parameters using the observation closest to where the peak would be. The right-most column is an estimate of the total mass swept up by the shock assuming a constant $n_e$.

**Table 9**. Optical Photometry

| Object | UTC Time | $t_{\rm rest}$ (days) | Facility | Filter | mag |
|---|---|---|---|---|---|
| AT2022abfc | 20221114 08:31 | −6 | P48/ZTF | $r$ | $> 19.68$ |
| AT2022abfc | 20221114 10:40 | −6 | P48/ZTF | $g$ | $> 19.65$ |
| AT2022abfc | 20221115 09:09 | −5 | P48/ZTF | $r$ | $> 20.40$ |
| AT2022abfc | 20221115 10:08 | −5 | P48/ZTF | $g$ | $> 20.19$ |
| AT2022abfc | 20221116 08:19 | −4 | P48/ZTF | $r$ | $> 19.36$ |
| AT2022abfc | 20221117 08:01 | −3 | P48/ZTF | $g$ | $> 19.69$ |
| AT2022abfc | 20221117 08:48 | −3 | P48/ZTF | $r$ | $> 19.53$ |
| AT2022abfc | 20221118 08:04 | −2 | P48/ZTF | $g$ | $> 20.51$ |
| AT2022abfc | 20221118 09:32 | −2 | P48/ZTF | $r$ | $> 20.13$ |
| AT2022abfc | 20221119 09:34 | −2 | P48/ZTF | $g$ | $> 19.71$ |
| AT2022abfc | 20221120 08:01 | −1 | P48/ZTF | $r$ | $> 20.35$ |
| AT2022abfc | 20221121 08:15 | 0 | P48/ZTF | $r$ | $19.46 \pm 0.11$ |
| AT2022abfc | 20221121 09:03 | 0 | P48/ZTF | $g$ | $19.49 \pm 0.14$ |
| AT2022abfc | 20221125 08:32 | 3 | P48/ZTF | $r$ | $19.55 \pm 0.16$ |
| AT2022abfc | 20221125 09:32 | 3 | P48/ZTF | $g$ | $19.71 \pm 0.20$ |
| AT2022abfc | 20221127 07:32 | 5 | P48/ZTF | $r$ | $20.32 \pm 0.32$ |
| AT2022abfc | 20221127 08:29 | 5 | P48/ZTF | $g$ | $20.21 \pm 0.19$ |
| AT2022abfc | 20221201 09:15 | 8 | P48/ZTF | $r$ | $> 20.23$ |
| AT2022abfc | 20221210 07:04 | 16 | P48/ZTF | $g$ | $> 19.60$ |
| AT2022abfc | 20221211 07:03 | 16 | P48/ZTF | $g$ | $> 19.85$ |
| AT2022abfc | 20221217 06:01 | 21 | P48/ZTF | $g$ | $> 20.35$ |
| AT2022abfc | 20221217 07:03 | 21 | P48/ZTF | $r$ | $> 20.19$ |
| AT2022abfc | 20221220 06:11 | 24 | P48/ZTF | $r$ | $> 19.21$ |
| AT2022abfc | 20221220 07:03 | 24 | P48/ZTF | $g$ | $> 19.66$ |
| AT2023fhn | 20230408 06:00 | −3 | P48/ZTF | $i$ | $> 19.54$ |
| AT2023fhn | 20230408 06:29 | −3 | P48/ZTF | $g$ | $> 18.76$ |
| AT2023fhn | 20230408 07:12 | −3 | P48/ZTF | $r$ | $> 19.17$ |
| AT2023fhn | 20230410 04:53 | −2 | P48/ZTF | $r$ | $19.68 \pm 0.09$ |
| AT2023fhn | 20230411 04:35 | −1 | P48/ZTF | $i$ | $19.65 \pm 0.1$ |
| AT2023fhn | 20230412 03:51 | 0 | P48/ZTF | $r$ | $19.13 \pm 0.05$ |
| AT2023fhn | 20230412 04:35 | 0 | P48/ZTF | $g$ | $18.69 \pm 0.03$ |
| AT2023fhn | 20230412 05:22 | 0 | P48/ZTF | $g$ | $18.63 \pm 0.03$ |
| AT2023fhn | 20230412 05:27 | 0 | P48/ZTF | $g$ | $18.72 \pm 0.03$ |
| AT2023fhn | 20230417 05:05 | 4 | P48/ZTF | $g$ | $20.09 \pm 0.09$ |
| AT2023fhn | 20230417 05:49 | 4 | P48/ZTF | $r$ | $20.72 \pm 0.15$ |
| AT2023fhn | 20230419 04:40 | 6 | P60/SEDM | $g$ | $> 19.30$ |
| AT2023fhn | 20230419 04:44 | 6 | P60/SEDM | $r$ | $> 19.93$ |
| AT2023fhn | 20230419 05:20 | 6 | P48/ZTF | $r$ | $20.57 \pm 0.22$ |

Note—**The full set of photometry for all six LFBOTs obtained from Fritz marshal are available online in a machine-readable table.** We omit all observations prior to six days before $t_0$ and all ZTF non-detections more than fifty days after $t_0$. For non-detections, we provide 3-$\sigma$ upper limits.